

# Beyond Accuracy: Explainable Trustworthy Index and Explainable Robustness Index for Evaluating CNN

Ragib Mehedi<sup>a,\*</sup>, Azmain Inquaid Haque<sup>a</sup>, Md. Nazmul Abdal<sup>b</sup>, Abdullah Al Noman<sup>a,\*</sup> and Md. Zahidul Islam<sup>a</sup>

<sup>a</sup>Computer Science and Engineering Discipline, Khulna University, Khulna, 9208, Bangladesh

<sup>b</sup>Department of Computer Science and Engineering, University of Liberal Arts Bangladesh, Dhaka, 1207, Bangladesh

## ARTICLE INFO

### Keywords:

Explainable AI (XAI), Trustworthiness, Robustness, CNN, Grad-CAM, Perturbation Analysis, Plant Leaf Classification

## ABSTRACT

Deep neural networks often achieve high classification accuracy. Despite high accuracy, the reliability of CNN models remains a concern. Sometimes CNN models may rely on background or irrelevant features, undermining trustworthiness. Besides, with the use of perturbations, models' accuracy and object reliance may decline. Such degradation raises questions about their robustness. In this research, we have proposed two novel evaluation metrics: the **Explainable Trustworthy Index (ETI)** and the **Explainable Robustness Index (ERI)**. ETI combines accuracy with the **Localization Mask Overlap Score (LMOS)**, which measures the proportion of Grad-CAM salient pixels that fall within the ground-truth object regions. A model with high ETI clearly indicates that both its accuracy and object reliance are high and satisfactory. ERI extends this to perturbed inputs by jointly evaluating accuracy stability and LMOS stability. Like ETI, a model with high ERI indicates that both its accuracy and object focus remain stable under the use of perturbations. We have used eight CNN architectures on four datasets and nine perturbations (e.g., noise, blur, motion blur, occlusion) to calculate ETI and ERI. We have demonstrated that high accuracy sometimes does not guarantee high ETI or ERI. For instance, on the EPLI dataset, Custom CNN has achieved 99.45% accuracy but the lowest ETI (0.5922) due to poor object localization. It has also provided the lowest ERI. ETI and ERI provide a more holistic assessment of model reliability for safety-critical applications.

## 1. Introduction

Convolutional Neural Networks (CNNs) have achieved remarkable performance in a wide range of domains such as medical imaging, agriculture, and astronomy, where predictions inform high-stakes decision-making processes [1, 2, 3, 4]. In such applications, classification accuracy is an essential performance measure, but accuracy alone does not necessarily indicate whether a model is reliable. A CNN model is considered more reliable when its predictions are grounded in task-relevant visual features rather than spurious correlations [5]. Therefore, beyond predictive accuracy, models must be quantitatively evaluated to determine whether their predictions depend on task-relevant regions rather than irrelevant background cues. Despite excellent accuracy, a model may not provide reliable predictions. Explainability methods, such as Grad-CAM [6], show that a model may rely mainly on background and irrelevant regions instead of focused objects [7, 8]. This raises questions about reliability and trustworthiness.

Model behavior can shift due to real-world perturbations e.g., *noise*, *blur*, *lighting shifts*, *compression*, and *occlusion*. These distortions can alter both CNN accuracy and the visual regions driving predictions. Consequently, even when a model maintains relatively high accuracy under perturbations, its underlying visual explanations may still become inconsistent [5, 9]. Conventional robustness evaluation methods are primarily based on accuracy change under perturbation. They do not capture whether the model maintains consistent reliance on task-relevant regions under perturbed conditions. These observations motivate us to develop an evaluation framework that considers both predictive behavior and explanation behavior.

Explainable Artificial Intelligence (XAI) provides a framework for examining visual evidences used by CNNs for their predictions. In particular, Grad-CAM generates class-discriminative heatmaps that highlight image regions contributing to a model's predictions [6, 10]. We refer to the Grad-CAM heatmaps as explanations corresponding to a model's predictions. Such explanations can reveal whether a model achieves high accuracy by focusing substantially on background or other irrelevant regions rather than the target objects. Previous studies have shown that background information and spurious correlations can influence deep neural network decisions and may lead to shortcut learning [11, 12, 13]. Consequently, a highly accurate model does not confirm that the predictions are made on appropriate object-level information. This creates an important distinction between predictive performance and explainable reliability. Motivated by these observations, we propose metrics that integrate accuracy with object focus.

\*Corresponding author

✉ ragib6627@gmail.com (R. Mehedi); azmaininquaidhaque@gmail.com (A.I. Haque); nazmul.abdal@ulab.edu.bd (Md.N. Abdal); noman@cse.ku.ac.bd (A.A. Noman); zahid@cse.ku.ac.bd (Md.Z. Islam)

🌐 <https://ulab.edu.bd/faculty/md-nazmul-abdal> (Md.N. Abdal); <https://ku.ac.bd/discipline/cse/faculty/noman> (A.A. Noman); <https://ku.ac.bd/discipline/cse/faculty/zahid> (Md.Z. Islam)

ORCID(s): 0009-0003-1555-7338 (R. Mehedi)

need a ref to all the methods mentioned below Prior literature has established diverse methodologies for evaluating the reliability, robustness, and explanation quality of convolutional neural networks (CNNs). Specifically, reliability has been assessed via uncertainty estimation [14], calibration confidence scoring, failure detection, and risk-coverage trade-offs [15], Robustness is typically evaluated by measuring performance degradation under common corruptions [16][17], natural distribution shifts [18], and targeted input perturbations [19, 20], with recent efforts also examining how corruptions affect explanation stability [21]. Concurrently, explanation quality and visual feature reliance have been measured using faithfulness metrics [22, 23], ground-truth alignment [], and the spatial overlap between visual explanations and target object regions [24].

Despite these advances, existing evaluation methods typically focus on isolated dimensions. Existing methods lack a unified, model-level evaluation framework that simultaneously accounts for predictive accuracy, the spatial fidelity of decision-relevant visual evidence, and their joint stability under input perturbations. A model can maintain high accuracy, or even demonstrate stable performance under data drifts, while relying on spuriously correlated visual cues. Conversely, its underlying visual focus may shift drastically even when accuracy appears invariant. To address these limitations, a comprehensive evaluation framework is needed to jointly assess predictive performance, object-centric explanation quality, and explanation-aware robustness.

We propose two novel evaluation metrics: Explainable Trustworthy Index (ETI) and Explainable Robustness Index (ERI) using XAI methods. Our objective is not to modify or improve CNN models, rather to provide quantitative measures for evaluating and comparing their reliability and robustness. ETI measures baseline reliability on unperturbed images by jointly evaluating classification accuracy and object-focused explanation quality. In contrast, ERI measures robustness to input perturbations by assessing whether accuracy and explanation quality remain consistent with clean-image behavior. In essence, ETI and ERI address two different questions: ETI asks “Whether a model is accurate and appropriately focused on the target object under normal conditions?”, whereas ERI asks “Whether these properties remain stable when the input is perturbed?”

Figure 1 illustrates the distinct advantages of our XAI-based metrics over traditional accuracy. Here, we show the predictions of two models - Xception [cite the Xception paper](#) and a CustomCNN (detailed in Section 5.3) - on the same test image. Both models were trained on identical training data for a leaf classification task, with each achieving an accuracy of 99.45% on the validation set. However, evaluating the models using our proposed metrics reveals that Xception achieves significantly higher ETI and ERI scores (0.88 and 0.93) compared to the custom CNN (0.59 and 0.79). Inspecting the Grad-CAM heatmaps confirms that the custom CNN relies heavily on background rather than the leaf object itself. While standard accuracy fails to identify this flaw, our

proposed metrics correctly indicate that selecting the custom CNN would lead to sub-optimal model deployment.

We introduce the Localization Mask Overlap Score (LMOS) to quantify object-focused explanation quality which is the proportion of the model’s decision-relevant activation that overlaps with the annotated target-object region. LMOS is used in both ETI and ERI calculation. It is observed that in Grad-CAM visualizations, red and yellow regions are activated most i.e., red and yellow have greater influence than other colors on models’ predictions [25, 26]. We integrate this principle into LMOS; a higher LMOS value indicates that a larger proportion of the model’s highlighted regions falls within the target object boundary rather than the background.

We comprehensively evaluate the proposed metrics using four real-world datasets comprising two leaf image datasets: EPLI [27], BDML[28] and two flower image datasets: FLRS [29], FR [30]. Eight CNN architectures are trained and evaluated on the datasets. We demonstrate that a custom CNN model achieves 99.45% classification accuracy on the EPLI dataset, yet its mean LMOS is only 0.1900, resulting in the lowest ETI score of 0.5922 among the evaluated models. Its LMOS is only 0.1900, indicating that only 19.00% of its decision-relevant activation, on average, falls within the annotated target-object regions, while the remaining 81.00% lies outside these regions. In contrast, the MobileNetV2 model achieves 99.45% accuracy (same as the custom CNN model) but obtains a substantially higher mean LMOS of 0.7765, indicating that 77.65% of its decision-relevant activation falls within the target-object regions, with only 22.35% outside them and an ETI of 0.8855. This demonstrates that models with similar high accuracy can exhibit substantially different levels of object-focused reliance and the model with higher LMOS should be selected for real-world deployment. Importantly, a low LMOS score does not inherently denote shortcut learning or overfitting, but rather quantifies the extent to which a model’s salient activations diverge from the target object.

The proposed ERI combines *accuracy stability* and *LMOS stability* to provide a joint measure of explainable robustness. This distinction is important because predictive stability and explanation stability are not correlated. For example, in our experiments we observe that the Custom CNN model under contrast perturbation maintains relatively high accuracy stability of 0.8494 but has substantially lower LMOS stability of 0.4545 on the EPLI dataset. This indicates that, although its predictive performance remains comparatively stable, its object-focused explanation changes substantially relative to the clean-image condition. Conversely, the ResNet50 model exhibits relatively low accuracy stability of 0.4599 but higher LMOS stability of 0.7830 under the same perturbation, indicating that its object-focused explanation remains comparatively more consistent despite a larger change in predictive performance. These demonstrate that accuracy stability alone cannot fully characterize robustness: a model may preserve its predictive performance while substantially changing its object-focused behavior, or it may

preserve its visual focus while its predictive performance deteriorates.

The significance of ETI and ERI lies in their ability to capture key dimensions of CNN reliability and robustness that standard accuracy metrics overlook. However, a high ETI or ERI score should not be equated with overall model trustworthiness, which encompasses broader factors beyond the scope of this work. Rather, these metrics focus specifically on predictive accuracy, object-focused explanation quality, and their stability under input perturbations - offering complementary criteria for evaluating CNN models beyond conventional accuracy benchmarks.

**Two questions that the proposed measures can answer - How trustworthy a model is? Trustworthy means - predictions are based on real objects in the images. How robust a model is? Robustness means - performance of the model will not degrade under perturbations.**

To show in exp:

- Acc high but not reliable - may be low robustness or predictions based on BG.
- a model having high acc but low LMOS and has one or more of the mentioned problems
- A model is trustworthy if it has high accuracy and high LMOS - show this. Idea: models with high ACC an high LMOS making more predictions based on objects in images rather than BGs
- Show LMOS still works with perturbations and ACC and other traditional measure fails.
- Justify distinctions between EI and RI
- Show - traditional measures fail under perturbations but proposed measures hold.

## 2. Problem Definition

Let  $D = \{(x_i, y_i, M_i)\}_{i=1}^N$  be a dataset containing  $N$  test images, where  $x_i \in \mathbb{R}^{n \times m}$  is the  $i$ -th image,  $y_i$  is its ground-truth label, and  $M_i$  is the corresponding ground-truth mask of the target object. Conventional evaluation methods primarily measure the classification accuracy of a model. Let  $acc \in [0, 1]$  denote the classification accuracy of a model.

Two CNNs may achieve identical accuracy while relying on substantially different image regions. In particular, a model may achieve high accuracy while its Grad-CAM activation map reveals that the activated regions concentrated outside the target object. Hence, accuracy is insufficient to distinguish models according to their object-focused explanatory behavior.

We seek to quantify the alignment between a model's rationale and its predictions. Let  $H_i$  denote the Grad-CAM heatmap resized to the input image corresponding to image  $x_i$  and prediction  $\hat{y}_i$ . We define a scoring function  $LMOS_i \in$

**Table 1**

Notation used in problem formulation.

| Notation      | Description                                                  |
|---------------|--------------------------------------------------------------|
| $D$           | Dataset of $N$ images with labels and target masks           |
| $N$           | Number of images in the dataset                              |
| $x_i$         | $i$ -th input image                                          |
| $y_i$         | Ground-truth class label of $x_i$                            |
| $\hat{y}_i$   | Predicted class label for $x_i$                              |
| $M_i$         | Ground-truth binary mask of the target object in $x_i$       |
| $H_i$         | Grad-CAM heatmap for $x_i$ and $\hat{y}_i$                   |
| $LMOS_i$      | Object-focused localization score for $x_i$                  |
| $LMOS$        | Mean LMOS over the evaluated dataset                         |
| $acc$         | Classification accuracy of a CNN                             |
| $\mathcal{P}$ | Set of $P$ input perturbations                               |
| $S_{acc}$     | Overall accuracy stability under perturbations               |
| $S_{LMOS}$    | Overall LMOS stability under perturbations                   |
| $T$           | Generic evaluation measure based on $acc$ and LMOS           |
| $R$           | Generic robustness measure based on $S_{acc}$ and $S_{LMOS}$ |
| $\Psi(\cdot)$ | Function combining $acc$ and LMOS                            |
| $\Phi(\cdot)$ | Function combining $S_{acc}$ and $S_{LMOS}$                  |

$[0, 1]$  to measure the overlap ratio between the decision-relevant regions of  $H_i$  and the ground-truth object mask  $M_i$ . To indicate model trustworthiness and robustness, we address the following two sub problems in this paper.

Firstly, we want to evaluate CNN models considering both their accuracy ( $acc$ ) and mean object-focused explanation quality (LMOS). Formally, the objective is to develop an evaluation measure  $T$  as a function of  $acc$  and LMOS:  $T = \Psi(acc, LMOS)$ . We address this problem by the proposed Explainable Trustworthy Index (ETI).

Secondly, we want to incorporate CNNs behavioural changes in response to perturbations in model evaluation metric. Let  $\mathcal{P} = \{p_1, p_2, \dots, p_P\}$  denote a set of input perturbations. A robust CNN should not only preserve its predictive performance under any changes in  $\mathcal{P}$  but also maintain a similar object-focused activation pattern similar to the model's behavior on the corresponding perturbed image. Let  $S_{acc}$  and  $S_{LMOS}$  represent the overall accuracy stability (the model's stability to perform under perturbation) and LMOS stability (the model's stability to remain the object-focused for making predictions) of a model. Our second problem is to obtain an evaluation measure  $R$  that jointly captures these two forms of stability across the evaluated perturbations:  $R = \Phi(S_{acc}, S_{LMOS})$ . We propose Explainable Robustness Index (ERI) to address this problem. Table 1 describes the notations used in the article. **Recheck - table has all the notations and same as used in the text**

### 3. System Overview

#### 4. Explainable Trustworthy Index and Explainable Robustness Index

Both of the proposed Explainable Trustworthy Index (ETI) and Explainable Robustness Index (ERI) metrics use Localization Mask Overlap Score (LMOS) that measures the consistency between model-generated explanations and ground-truth object regions. ETI assesses if predictions are grounded in actual object regions rather than background noise. ERI extends this framework to perturbed inputs, measuring joint predictive and localization stability.

##### 4.1. Localization Mask Overlap Score (LMOS)

LMOS calculates how much a model focuses on true object regions (instead of backgrounds) for prediction. The Grad-CAM heatmaps are generated from the last convolutional layer of a network. They are resized to the input image size. Let  $H \in \mathbb{R}^{h \times w}$  denote the resized heatmap. We obtain a binary attention map  $\hat{H}(i, j)$  from  $H$  by thresholding:

$$\hat{H}(i, j) = \mathbb{1}(H(i, j) > \tau) \quad (1)$$

where  $\mathbb{1}(\cdot)$  is an indicator function and  $\tau$  is the threshold. We set  $\tau = 0.7$  (see Section 5.4) to restrict our LMOS calculations to red and yellow pixels, as the red and yellow pixels exhibit the highest activation levels in model predictions. Hence,  $\hat{H}(i, j)$  is a binary mask representing the activated decision-relevant regions for the corresponding predictions and input image.

We define LMOS as the ratio of activated decision-relevant regions inside the object region and the total activated decision-relevant regions. Mathematically:

$$\text{LMOS} = \frac{\sum_{i=1}^h \sum_{j=1}^w \hat{H}(i, j) M(i, j)}{\sum_{i=1}^h \sum_{j=1}^w \hat{H}(i, j)} \quad (2)$$

where  $M \in \{0, 1\}^{h \times w}$  denote the ground-truth binary segmentation mask.

##### 4.2. Explainable Trustworthy Index (ETI)

We propose Explainable Trustworthy Index (ETI) to measure trustworthiness of a CNN model's predictions. We combine the accuracy, explainable consistency and the mean LMOS (Eq. 2) to define ETI. Note that the LMOS is an instance specific (or local) measure. On the other hand accuracy is a global measure indicating the overall performance of a model.

Let  $acc$  denote the classification accuracy of a model on test set and  $\overline{\text{LMOS}}$  denote the mean LMOS scores across the same test samples. We define ETI as follows:

$$\text{ETI} = \frac{acc + \overline{\text{LMOS}}}{2}. \quad (3)$$

In this study, both  $acc$  and LMOS are equally important and are normalized to the range  $[0, 1]$ . Here,  $acc$  measures the overall predictive performance of a model, whereas

LMOS measures the mean explanation consistency specific to the target object. We assign equal weighting to both components to avoid favoring either predictive performance or explanatory quality.

##### 4.3. Explainable Robustness Index (ERI)

We estimate the robustness of a model by measuring its performance deviation under perturbation. Let  $acc_c$  be the accuracy of a model on a clean dataset (not perturbed) and  $acc_p$  denote the accuracy obtained on the same dataset after applying a specific perturbation  $p$  (see Sec. 5.2 for all the perturbation methods used in this research). We introduce the Accuracy Stability ( $S_{acc}$ ) score **if not introduced by you cite** under a specific perturbation  $p$  as:

$$S_{acc}^p = 1 - \frac{|acc_c - acc_p|}{acc_c + acc_p + \epsilon}$$

where  $\epsilon$  is a small positive constant introduced to avoid division by zero error. A higher  $S_{acc}^p$  value indicates greater performance consistency under the perturbed condition  $p$ .

To estimate an overall measure of accuracy stability ( $S_{acc}$ ), we compute  $S_{acc}^p$  across all  $p \in \mathcal{P}$  where  $\mathcal{P}$  is a set of perturbation methods, see Section 5.2, and take the mean of all  $S_{acc}^p$ . We define this as follows:

$$S_{acc} = \frac{1}{|\mathcal{P}|} \sum_{p=1}^{|\mathcal{P}|} S_{acc}^p$$

Similarly, we measure LMOS stability ( $S_{LMOS}$ ) to quantify how consistently a model maintains its visual object focus under input perturbations relative to unperturbed images. For each sample in a dataset, we compute LMOS for both the original image and its perturbed version using Eq. 2.

Let  $\text{LMOS}_c^i$  be the LMOS value computed from sample  $i$  and  $\text{LMOS}_p^i$  be the LMOS value computed from the same sample  $i$  after applying perturbation  $p$ . To quantify the consistency between these two values, we define the LMOS stability under a perturbation  $p$  for sample  $i$  as:

$$S_{LMOS}^{p,i} = 1 - \frac{|\text{LMOS}_c^i - \text{LMOS}_p^i|}{\text{LMOS}_c^i + \text{LMOS}_p^i + \epsilon}$$

where  $\epsilon$  is a small constant to avoid division by zero error. A higher value of  $S_{LMOS}^{p,i}$  indicates that the model's reliance on the focus object remains more consistent under the corresponding perturbation.

Given there are  $N$  test images in a dataset, the LMOS stability under the perturbation  $p$  across the entire dataset is estimated as:

$$S_{LMOS}^p = \frac{1}{N} \sum_{i=1}^N S_{LMOS}^{p,i}.$$

To obtain an overall measure of LMOS stability across all evaluated perturbations  $\mathcal{P}$ , the perturbation-specific LMOS stability scores ( $S_{LMOS}^p$ ) are subsequently averaged as:

$$S_{LMOS} = \frac{1}{|\mathcal{P}|} \sum_{p=1}^{|\mathcal{P}|} S_{LMOS}^p.$$

To jointly evaluate the robustness of predictive accuracy and explainability under input perturbations, we propose the *Explainable Robustness Index* (ERI). While accuracy stability and LMOS stability independently assess predictive consistency and visual localization behavior, ERI provides a unified robustness measure by integrating both dimensions.

The ERI under perturbation  $p$  ( $ERI^p$ ) is then defined by combining the accuracy stability  $S_{acc}^p$  and LMOS stability  $S_{LMOS}^p$  under perturbation  $p$  as follows:

$$ERI^p = \frac{S_{acc}^p + S_{LMOS}^p}{2}. \quad (4)$$

The overall ERI is defined as the mean of overall accuracy stability and overall consistency of model's reliance on object focus as follows:

$$ERI = \frac{S_{acc} + S_{LMOS}}{2}. \quad (5)$$

Here,  $S_{acc}$  represents the overall consistency of classification accuracy between clean and perturbed inputs, while  $S_{LMOS}$  represents the overall consistency of the model's reliance on the focus object under perturbations. ERI jointly captures two complementary dimensions of robustness: *predictive stability* and *explanatory stability*.

Since both constituent scores share a common upper and lower bound, the resulting ERI serves as a normalized metric of overall robustness. A higher ERI value signifies that a model concurrently preserves both its classification accuracy and object-centric visual focus across input perturbations. Conversely, a lower ERI reflects marked instability in either predictive performance, visual focus, or both.

## 5. Experimental Setup

### 5.1. Datasets

We have used four publicly available datasets in our empirical study of the proposed evaluation measures:

- **EPLI**: EPLI (introduced as EgyPLI in [27]) contains 3,588 JPG images of size  $256 \times 256$  pixels. There are images of 8 species including apple, berry, fig, guava, orange, plum, persimmon, and tomato. The dataset is annotated for image-level classification [27].
- **BDML**: BDML(also known as BDMediLeaves [28]) is a dataset of 2,029 original leaf images (augmented to 38,606) from 10 medicinal plant species. They are captured with an iPhone 13 under natural light at  $3024 \times 4032$  resolution. This dataset is designed for machine learning-based species identification. We used the original dataset (not the augmented one) in our experiments [28].
- **FLRS**: FLRS (or Flowers [29]) is a large image dataset for flower classification containing 3,670 training images of five flower species including daisies,

tulips, sunflowers, dandelions, and roses. It is available through the TensorFlow Datasets (TFDS) collection [29]. [what is the resolution of the images in this dataset?](#)

- **FR**: This kaggle [30] dataset is very similar to the FLRS dataset above consisting the same five species (daisies, dandelions, roses, sunflowers, and tulips). However, the kaggle version contains 4,242 JPG images of resolution of  $320 \times 240$  pixels. Although the image classes are the same as FLRS, the image samples are different. This dataset is used in previous research [31, 32].

**Dataset preparation.** The BDML dataset is already splitted into training, validation and testing sets. We split the other three datasets into training, validation and test subsets with an approximate ratio of 80%, 10% and 10%. Table 2 presents the class-wise training, validation, and test splits. [did you preserve the class distributions in the three splits?](#)

[did you do this? or already in the sources?](#) To simplify explainability and localization analysis, the test images have been manually annotated to generate corresponding JSON files. These JSON annotations describe the polygonal boundaries of the target leaf in each image. The JSON files have been converted into binary masks. Here, each mask represents only the target leaf region and excludes the background and other objects in the scene. [Table is missing BDML distribution](#)

### 5.2. Perturbation methods

To evaluate the robustness of the proposed framework under distribution shifts and common image corruptions, we have applied nine image perturbation methods to the test set. [if anything in the table is missing below, add it in the corresponding bullet](#) The perturbation parameters have been fixed across all experiments to ensure reproducibility and enable fair comparisons among different CNN architectures. The details of the perturbations are given below: [missing references here. Add 1 or more references to papers applying these for the same reason? i.e., evaluation under perturbation](#)

- **Gaussian Blur**: Applied using a  $5 \times 5$  kernel to simulate the loss of fine spatial details caused by defocus or low-quality imaging conditions.
- **Gaussian Noise**: Additive Gaussian noise with zero mean and standard deviation 0.05 has been introduced to model sensor noise and random intensity fluctuations.
- **Brightness Adjustment**: Increased by adding a constant value of 0.2 to normalized pixel intensities, after which values have been clipped to the valid range [0, 1].
- **Hue Shift**: Hue values in the HSV color space have been shifted by 15 units while preserving the saturation and value channels, simulating color variations caused by illumination or camera characteristics.

**Table 2**

Class-wise distribution of the experimental datasets.

| Dataset / Class | Total        | Train        | Val        | Test       |
|-----------------|--------------|--------------|------------|------------|
| <i>EPLI</i>     |              |              |            |            |
| Apple           | 519          | 415          | 52         | 52         |
| Berry           | 340          | 272          | 34         | 34         |
| Fig             | 508          | 406          | 50         | 52         |
| Guava           | 520          | 416          | 52         | 52         |
| Orange          | 547          | 437          | 55         | 55         |
| Plum            | 468          | 374          | 47         | 47         |
| Persimmon       | 527          | 421          | 53         | 53         |
| Tomato          | 159          | 127          | 16         | 16         |
| <b>Subtotal</b> | <b>3,588</b> | <b>2,868</b> | <b>359</b> | <b>361</b> |
| <i>FIRS</i>     |              |              |            |            |
| Daisy           | 633          | 506          | 64         | 63         |
| Dandelion       | 898          | 718          | 90         | 90         |
| Roses           | 641          | 512          | 64         | 65         |
| Sunflowers      | 699          | 560          | 70         | 69         |
| Tulips          | 799          | 639          | 80         | 80         |
| <b>Subtotal</b> | <b>3,670</b> | <b>2,935</b> | <b>368</b> | <b>367</b> |
| <i>FR</i>       |              |              |            |            |
| Daisy           | 764          | 611          | 76         | 77         |
| Dandelion       | 1052         | 842          | 105        | 105        |
| Roses           | 784          | 627          | 78         | 79         |
| Sunflowers      | 733          | 586          | 73         | 74         |
| Tulips          | 984          | 787          | 98         | 99         |
| <b>Subtotal</b> | <b>4,317</b> | <b>3,453</b> | <b>430</b> | <b>434</b> |

- **Contrast Adjustment:** Modified using mean-centered linear scaling:  $I' = (I - \mu_I) \times 0.5 + \mu_I$ , where  $I$  denotes the input image and  $\mu_I$  represents the mean pixel intensity (computed per channel). The resulting values have been clipped to the range  $[0, 1]$ .
- **Gamma Correction:** Nonlinear illumination changes have been simulated using gamma correction:  $I' = I^{2.0}$ , where pixel intensities have been raised to the power of  $\gamma = 2.0$ .
- **Motion Blur:** Simulated using a  $7 \times 7$  horizontal averaging kernel to emulate camera or object motion during image acquisition.
- **JPEG Compression:** Artifacts have been introduced by encoding and decoding the images with a quality factor of 30, simulating degradation due to lossy storage or transmission.
- **Random Occlusion:** A randomly positioned rectangular region corresponding to 20% of the image height and 20% of the image width has been masked by setting its pixel values to zero, thereby simulating partial object occlusion.

### 5.3. Implementaion the CNN models

In our experiments, we have used the following CNN models:

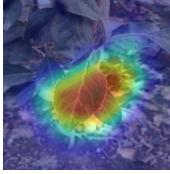
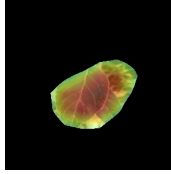
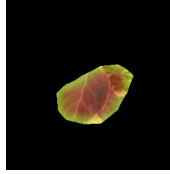
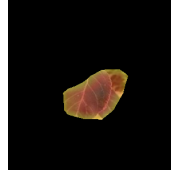
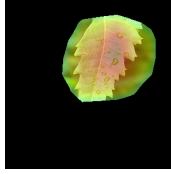
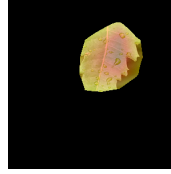
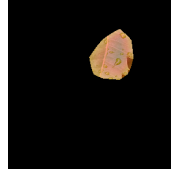
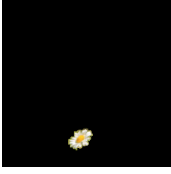
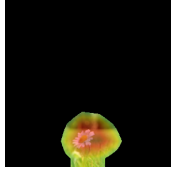
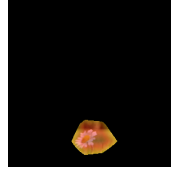
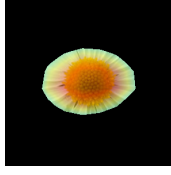
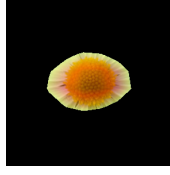
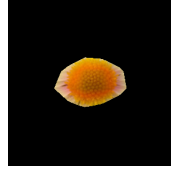
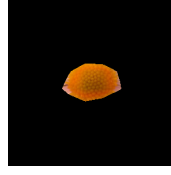
- **MobileNetV2:** MobileNetV2 is a lightweight CNN architecture designed to achieve a favorable trade-off between classification performance and computational efficiency. It employs inverted residual blocks with linear bottlenecks and lightweight depthwise convolutions to reduce computational cost while maintaining effective feature representations[33].
- **Xception:** Xception is based on depthwise separable convolutions and can be viewed as an extension of the Inception design in which spatial and channel-wise feature learning are more explicitly separated. The architecture provides an efficient use of model parameters while maintaining strong image classification performance cite r201.
- **DenseNet121:** DenseNet121 introduces dense connectivity, in which each layer receives feature maps from all preceding layers within a dense block. This design encourages feature reuse, strengthens feature propagation, and helps alleviate the vanishing-gradient problem while maintaining relatively efficient parameter usage [34].
- **InceptionResNetV2:** InceptionResNetV2 combines the multi-branch Inception architecture with residual connections to facilitate the training of deep networks. The residual connections accelerate optimization and enable effective learning in a computationally efficient Inception-based architecture [35].
- **VGG16:** VGG16 is a deep convolutional architecture that uses a simple and uniform design based primarily on stacked  $3 \times 3$  convolutional filters followed by pooling layers. Its increased depth and straightforward architecture have made it a widely used baseline for image classification and transfer learning [36]
- **InceptionV3:** InceptionV3 employs factorized convolutions and carefully designed Inception modules to increase representational capacity while controlling computational cost. The architecture also incorporates auxiliary classification and other design improvements to achieve effective image recognition performance [37].
- **ResNet50:** ResNet50 employs residual learning through shortcut connections that allow layers to learn residual mappings with respect to their inputs. This design facilitates the optimization of deeper networks and enables the effective learning of hierarchical visual representations [38].
- **CustomCNN:** In this research, it has been developed as the original EPLI research.

### 5.4. Implementaion of ETI and ERI

Use used the Grad-CAM implementation from `specify[]`. We apply Grad-CAM to obtain the heatmap  $H$  for output of each model (described above in Section 5.3). We resized

**Table 3**

Comparison of Grad-CAM masked visualizations under different threshold values.

| Original Image                                                                    | Masked Image                                                                      | Grad-CAM Visualization                                                            | Threshold 0.5                                                                     | Threshold 0.6                                                                      | Threshold 0.7                                                                       | Threshold 0.8                                                                       |
|-----------------------------------------------------------------------------------|-----------------------------------------------------------------------------------|-----------------------------------------------------------------------------------|-----------------------------------------------------------------------------------|------------------------------------------------------------------------------------|-------------------------------------------------------------------------------------|-------------------------------------------------------------------------------------|
|  |  |  |  |  |  |  |
|  |  |  |  |  |  |  |
|  |  |  |  |  |  |  |
|  |  |  |  |  |  |  |

the heatmaps to the corresponding model's input image size, e.g., for Xception it is  $299 \times 299$  and for other models, it is  $224 \times 224$ .

The next step in our method is to get the binary attention map  $\hat{H}(i, j)$  for each sample image (see Eq. 1). We select the value of threshold  $\tau$  through experiments are discussed below.

Using the binarized heatmap  $\hat{H}(i, j)$  and the ground-truth binary mask  $M(i, j)$ , we compute LMOS according to Eq. 2. We created the ground-truth binary masks from the corresponding JSON files of test images and resized into the input image size (similar to the heatmaps). This same LMOS computation procedure is subsequently used to derive the proposed ETI and ERI scores, as defined in Eq. 3 and Eq. 5, respectively.

**Selection of the threshold  $\tau$ .** The choice of the threshold parameter  $\tau$  is critical, as it governs the filtering of decision-relevant regions during heatmap binarization. We experimented with  $\tau \in \{0.5, 0.6, 0.7, 0.8\}$ . Results of the experiments are shown in Table 3. *need to work on the figures*

At  $\tau = 0.5$ , the binarized mask retains a broad portion of the visual attribution map capturing strong activity in red and yellow regions. It also illustrates substantial weak activity in the green regions. While this lower threshold preserves more total activations, incorporating these peripheral regions risks introducing irrelevant visual evidence into the masked Grad-CAM. Consequently, this raises concerns regarding whether

the resulting binary mask sufficiently isolates true decision-relevant regions.

Increasing  $\tau = 0.6$  improves selectivity by suppressing most green activations, yet it permits weakly activated peripheral regions to pass alongside core object features. Conversely,  $\tau = 0.8$  imposes overly stringent constraints, retaining predominantly red regions while discarding valid, decision-relevant yellow activations.

As illustrated in Table 3,  $\tau = 0.7$  achieves an optimal balance by isolating the primary red and yellow heatmaps (representing the model's core decision-relevant evidence) while successfully suppressing low-intensity background noise. Based on this experiment, we select  $\tau = 0.7$  for  $\hat{H}(i, j)$  computation and applied consistently across all datasets, architectures, and perturbations to ensure cross-experimental comparability.

**Variations of ETI.** To demonstrate the impact of LMOS and  $acc$  on the proposed ETI, we have calculated two variants of ETI as follows:

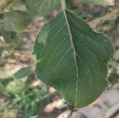
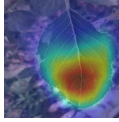
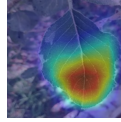
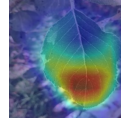
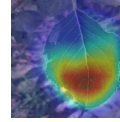
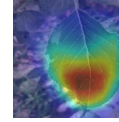
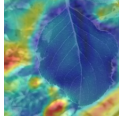
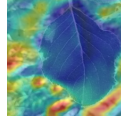
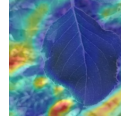
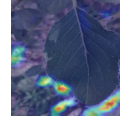
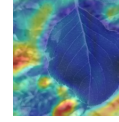
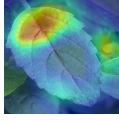
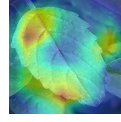
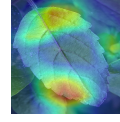
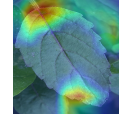
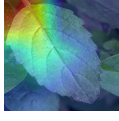
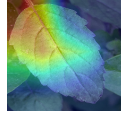
$$ETI_{w/o \text{ LMOS}} = \frac{acc}{2}, \quad ETI_{w/o \text{ acc}} = \frac{\overline{LMOS}}{2}.$$

**Variations of ERI.** To analyze the contribution of the individual components of ERI (i.e.,  $S_{acc}$  and  $S_{LMOS}$ ), we calculate two variants by removing one component at a time:

$$ERI_{w/o \text{ LMOS}} = \frac{S_{acc}}{2}, \quad ERI_{w/o \text{ acc}} = \frac{S_{LMOS}}{2}.$$

## 5.5. Evaluation metrics

**Table 4**  
Grad-CAM visualization on EPLI and BDML datasets under clean and perturbed conditions.

| Model Metrics                                                                                                                 | Original                                                                          | True Mask (M)                                                                     | Clean                                                                             | Hue                                                                               | Blur                                                                                | Contrast                                                                            | Compression                                                                         |
|-------------------------------------------------------------------------------------------------------------------------------|-----------------------------------------------------------------------------------|-----------------------------------------------------------------------------------|-----------------------------------------------------------------------------------|-----------------------------------------------------------------------------------|-------------------------------------------------------------------------------------|-------------------------------------------------------------------------------------|-------------------------------------------------------------------------------------|
| <b>Model:</b> Xception<br><b>Accuracy:</b> 99.45%<br><b>Mean LMOS:</b> 0.7638<br><b>ETI:</b> 0.8791<br><b>ERI:</b> 0.9346     |  |  |  |  |  |  |  |
|                                                                                                                               | Apple (EPLI)                                                                      |                                                                                   | LMOS: 1.00                                                                        | LMOS: 1.00                                                                        | LMOS: 1.00                                                                          | LMOS: 1.00                                                                          | LMOS: 1.00                                                                          |
| <b>Model:</b> Custom CNN<br><b>Accuracy:</b> 99.45%<br><b>Mean LMOS:</b> 0.1900<br><b>ETI:</b> 0.5922<br><b>ERI:</b> 0.7944   |  |  |  |  |  |  |  |
|                                                                                                                               | Apple (EPLI)                                                                      |                                                                                   | LMOS: 0.00                                                                        | LMOS: 0.00                                                                        | LMOS: 0.00                                                                          | LMOS: 0.00                                                                          | LMOS: 0.00                                                                          |
| <b>Model:</b> Xception<br><b>Accuracy:</b> 95.05%<br><b>Mean LMOS:</b> 0.7800<br><b>ETI:</b> 0.8653<br><b>ERI:</b> 0.8982     |  |  |  |  |  |  |  |
|                                                                                                                               | O. tenuiflorum (BDML)                                                             |                                                                                   | LMOS: 1.00                                                                        | LMOS: 1.00                                                                        | LMOS: 0.903                                                                         | LMOS: 0.812                                                                         | LMOS: 0.785                                                                         |
| <b>Model:</b> Inception V3<br><b>Accuracy:</b> 93.07%<br><b>Mean LMOS:</b> 0.3344<br><b>ETI:</b> 0.6325<br><b>ERI:</b> 0.7898 |  |  |  |  |  |  |  |
|                                                                                                                               | O. tenuiflorum (BDML)                                                             |                                                                                   | LMOS: 0.383                                                                       | LMOS: 0.659                                                                       | LMOS: 0.477                                                                         | LMOS: 0.390*                                                                        | LMOS: 0.176                                                                         |

Write about the accuracy metric here and the Rank column in the tables.

## 5.6. Training Configurations

We used TensorFlow/Keras on the Kaggle [cite kaggle](#) environment with two NVIDIA Tesla T4 GPUs to training and evaluation of all the CNN models in our experiments. The training configuration has followed the settings adopted in the original EPLI study[10]. [what are these? please specify. is it only for EgyPLI or all 4 datasets??](#) The custom CNN was trained from scratch, while other models have been initialized with ImageNet-pretrained weights [cite here](#) and their backbone layers have been frozen.

For the pretrained models, the original classification heads have been removed and replaced with a common classification head consisting of Global Average Pooling [cite](#), a dropout layer [cite](#) with dropout rate 0.3, a fully connected layer with 128 ReLU units [cite](#), and the final softmax layer has been configured according to the number of classes in each dataset. The Adam optimiser [cite](#) has been used with a learning rate of  $5 \times 10^{-5}$  and categorical cross-entropy loss [arg1](#) in all model optimizations.

## 6. Result and Discussion

### 6.1. Grad-CAM Visualization

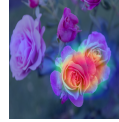
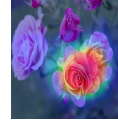
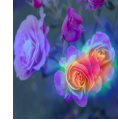
Tables 4 and 5 present representative Grad-CAM visualizations with corresponding LMOS values across the four benchmark datasets. For each selected sample, the visualization strip contains the original input image, its ground-truth segmentation mask, the segmented target obtained by removing the background, and Grad-CAM visualizations

under clean and perturbed conditions. These qualitative examples complement the quantitative ETI and ERI results by illustrating how different CNN architectures focus on the target object and how their attention patterns change under input perturbations.

On the EPLI dataset, Xception achieves a high mean LMOS of 0.7638 and an ETI of 0.8791, indicating that its classification decisions are generally associated with regions corresponding to the target leaf. As shown in Table 4, the Grad-CAM maps remain well aligned with the target region across the clean, hue, blur, contrast, and compression conditions, with an LMOS of 1.00 for the selected sample. In contrast, the Custom CNN achieves the same classification accuracy of 99.45% but obtains a substantially lower mean LMOS of 0.1900 and an ETI of 0.5922. Its representative Grad-CAM maps show an LMOS of 0.00 across the displayed conditions, demonstrating that high classification accuracy alone does not necessarily imply meaningful object-focused attention.

A similar pattern is observed on BDML. Xception obtains a mean LMOS of 0.7800 and an ETI of 0.8653, while InceptionV3 obtains considerably lower values of 0.3344 and 0.6325, respectively. For the selected BDML sample, Xception maintains strong localization under the clean and hue conditions, while its attention becomes less concentrated under blur, contrast, and compression, with LMOS values of 0.903, 0.812, and 0.785, respectively. InceptionV3 exhibits substantially weaker localization, with LMOS values ranging from 0.176 to 0.659 across the displayed conditions. This qualitative difference is consistent with the corresponding ETI values and demonstrates the ability of LMOS to

**Table 5**  
Grad-CAM visualization on Flower datasets (FLRS and FR).

| Model Metrics                                                                                                                 | Original                                                                          | True Mask (M)                                                                     | Clean                                                                             | Hue                                                                               | Blur                                                                                | Contrast                                                                            | Compression                                                                         |
|-------------------------------------------------------------------------------------------------------------------------------|-----------------------------------------------------------------------------------|-----------------------------------------------------------------------------------|-----------------------------------------------------------------------------------|-----------------------------------------------------------------------------------|-------------------------------------------------------------------------------------|-------------------------------------------------------------------------------------|-------------------------------------------------------------------------------------|
| <b>Model:</b> Xception<br><b>Accuracy:</b> 88.28%<br><b>Mean LMOS:</b> 0.6869<br><b>ETI:</b> 0.7849<br><b>ERI:</b> 0.9308     |  |  |  |  |  |  |  |
|                                                                                                                               | Rose (FLRS)                                                                       |                                                                                   | LMOS: 1.00                                                                        | LMOS: 1.00                                                                        | LMOS: 1.00                                                                          | LMOS: 1.00                                                                          | LMOS: 1.00                                                                          |
| <b>Model:</b> Inception V3<br><b>Accuracy:</b> 82.29%<br><b>Mean LMOS:</b> 0.3749<br><b>ETI:</b> 0.5989<br><b>ERI:</b> 0.8593 |  |  |  |  |  |  |  |
|                                                                                                                               | Rose (FLRS)                                                                       |                                                                                   | LMOS: 0.305                                                                       | LMOS: 0.219                                                                       | LMOS: 0.181                                                                         | LMOS: 0.219                                                                         | LMOS: 0.214                                                                         |
| <b>Model:</b> Xception<br><b>Accuracy:</b> 90.32%<br><b>Mean LMOS:</b> 0.6029<br><b>ETI:</b> 0.7531<br><b>ERI:</b> 0.9689     |  |  |  |  |  |  |  |
|                                                                                                                               | Daisy (FR)                                                                        |                                                                                   | LMOS: 1.00                                                                        | LMOS: 1.00                                                                        | LMOS: 1.00                                                                          | LMOS: 1.00                                                                          | LMOS: 1.00                                                                          |
| <b>Model:</b> Inception V3<br><b>Accuracy:</b> 88.25%<br><b>Mean LMOS:</b> 0.3063<br><b>ETI:</b> 0.5944<br><b>ERI:</b> 0.8827 |  |  |  |  |  |  |  |
|                                                                                                                               | Daisy (FR)                                                                        |                                                                                   | LMOS: 0.633                                                                       | LMOS: 0.583                                                                       | LMOS: 0.550                                                                         | LMOS: 0.388                                                                         | LMOS: 0.505                                                                         |

distinguish models that may otherwise exhibit relatively competitive classification performance.

The flower datasets further demonstrate that the relationship between classification accuracy and explanation quality is not uniform across datasets. On FLRS, Xception achieves an ETI of 0.7849 with a mean LMOS of 0.6869, whereas InceptionV3 obtains an ETI of 0.5989 and a mean LMOS of 0.3749. The selected FLRS sample shows that Xception maintains an LMOS of 1.00 across all displayed perturbations, whereas the InceptionV3 maps exhibit substantially lower localization scores, ranging from 0.181 to 0.305 for several perturbations. On FR, Xception again demonstrates stronger explanation quality, achieving a mean LMOS of 0.6029 and an ETI of 0.7531. Its selected sample maintains an LMOS of 1.00 under all displayed conditions, while InceptionV3 produces lower and more variable localization scores, ranging from 0.388 to 0.633.

Overall, these visualizations reinforce the main motivation of ETI: classification accuracy alone is insufficient to characterize the reliability of a CNN when the model may base its prediction on regions unrelated to the target object. Models such as Xception demonstrate that strong classification performance can be accompanied by substantially better target-focused attention, whereas the Custom CNN and, in several cases, InceptionV3 exhibit a noticeable discrepancy between predictive performance and localization quality. The qualitative Grad-CAM results therefore provide an intuitive interpretation of the quantitative LMOS and ETI measurements and illustrate how the proposed evaluation framework can expose differences in model behavior that conventional accuracy metrics may overlook.

## 6.2. Explainable Trustworthiness Index (ETI)

Table 6 illustrates a detailed evaluation of the proposed Explainable Trustworthiness Index (ETI) across four benchmark datasets: EPLI, BDML, FLRS, and FR. ETI not only considers accuracy but also explanation quality. It integrates prediction accuracy and the mean Localization Mask Overlap Score (LMOS). Therefore, a model achieves a higher ETI only when it possesses both reliable classification capability and consistent focus on semantically meaningful image regions instead of the background. The results clearly indicate that accuracy alone is not enough to measure model trustworthiness. Several models have achieved competitive classification performance but fail to achieve higher ETI values due to poor localization quality.

For the EPLI dataset, MobileNetV2 and Xception have achieved the highest ETI values of 0.8855 and 0.8791, respectively, despite DenseNet121 obtaining the highest classification accuracy of 1.0000. Although DenseNet121 perfectly classifies all test samples, its lower LMOS value (0.6964) reduces its overall ETI to 0.8482. This observation highlights the importance of explanation consistency as well as predictive accuracy for trustworthiness. Similarly, the Custom CNN achieves a high accuracy of 0.9945, which is comparable to the best-performing architectures. However, its LMOS value is the lowest (0.1900). So, ultimately, the ETI value is the lowest (0.5922). This indicates that the model may rely on irrelevant image regions rather than the focus leaf regions for decision-making.

A similar trend is observed on the BDML dataset. MobileNetV2 and Xception provide the highest ETI scores

**Table 6**

Comparison of classification accuracy, mean LMOS, ETI, and ablation variants across different CNN architectures on four benchmark datasets.

| Dataset | Rank | Model             | Acc    | LMOS   | ETI    | ETI w/o Acc. | ETI w/o LMOS |
|---------|------|-------------------|--------|--------|--------|--------------|--------------|
| EPLI    | 1    | MobileNetV2       | 0.9945 | 0.7765 | 0.8855 | 0.3882       | 0.4972       |
|         | 2    | Xception          | 0.9945 | 0.7638 | 0.8791 | 0.3819       | 0.4972       |
|         | 3    | DenseNet121       | 1.0000 | 0.6964 | 0.8482 | 0.3482       | 0.5000       |
|         | 4    | InceptionResNetV2 | 0.9889 | 0.5529 | 0.7709 | 0.2764       | 0.4945       |
|         | 5    | VGG16             | 0.9834 | 0.5268 | 0.7551 | 0.2634       | 0.4917       |
|         | 6    | InceptionV3       | 0.9917 | 0.3617 | 0.6767 | 0.1809       | 0.4958       |
|         | 7    | ResNet50          | 0.6122 | 0.7256 | 0.6689 | 0.3628       | 0.3061       |
|         | 8    | Custom CNN        | 0.9945 | 0.1900 | 0.5922 | 0.0950       | 0.4972       |
| BDML    | 1    | MobileNetV2       | 0.9703 | 0.7611 | 0.8657 | 0.3805       | 0.4851       |
|         | 2    | Xception          | 0.9505 | 0.7800 | 0.8653 | 0.3900       | 0.4752       |
|         | 3    | DenseNet121       | 0.9802 | 0.7118 | 0.8460 | 0.3559       | 0.4901       |
|         | 4    | VGG16             | 0.8960 | 0.5873 | 0.7416 | 0.2936       | 0.4480       |
|         | 5    | InceptionResNetV2 | 0.9059 | 0.5184 | 0.7122 | 0.2592       | 0.4530       |
|         | 6    | InceptionV3       | 0.9307 | 0.3344 | 0.6325 | 0.1672       | 0.4653       |
|         | 7    | ResNet50          | 0.4604 | 0.7817 | 0.6211 | 0.3909       | 0.2302       |
|         | 8    | Custom CNN        | 0.7376 | 0.3310 | 0.5343 | 0.1655       | 0.3688       |
| FLRS    | 1    | Xception          | 0.8828 | 0.6869 | 0.7849 | 0.3435       | 0.4414       |
|         | 2    | DenseNet121       | 0.9046 | 0.6188 | 0.7617 | 0.3094       | 0.4523       |
|         | 3    | MobileNetV2       | 0.8665 | 0.5958 | 0.7311 | 0.2979       | 0.4332       |
|         | 4    | VGG16             | 0.8147 | 0.6311 | 0.7229 | 0.3155       | 0.4074       |
|         | 5    | InceptionResNetV2 | 0.8501 | 0.5581 | 0.7041 | 0.2791       | 0.4251       |
|         | 6    | InceptionV3       | 0.8229 | 0.3749 | 0.5989 | 0.1875       | 0.4114       |
|         | 7    | ResNet50          | 0.5123 | 0.5623 | 0.5373 | 0.2812       | 0.2561       |
|         | 8    | Custom CNN        | 0.7003 | 0.2891 | 0.4947 | 0.1446       | 0.3501       |
| FR      | 1    | Xception          | 0.9032 | 0.6029 | 0.7531 | 0.3015       | 0.4516       |
|         | 2    | DenseNet121       | 0.9124 | 0.5382 | 0.7253 | 0.2691       | 0.4562       |
|         | 3    | VGG16             | 0.8387 | 0.5786 | 0.7087 | 0.2893       | 0.4194       |
|         | 4    | MobileNetV2       | 0.8986 | 0.5091 | 0.7038 | 0.2545       | 0.4493       |
|         | 5    | InceptionResNetV2 | 0.8894 | 0.4705 | 0.6800 | 0.2353       | 0.4447       |
|         | 6    | InceptionV3       | 0.8825 | 0.3063 | 0.5944 | 0.1531       | 0.4412       |
|         | 7    | ResNet50          | 0.4977 | 0.4769 | 0.4873 | 0.2384       | 0.2488       |
|         | 8    | Custom CNN        | 0.6820 | 0.2714 | 0.4767 | 0.1357       | 0.3410       |

of 0.8657 and 0.8653, respectively, showing balanced classification capability and explanation reliability. In contrast, DenseNet121 achieves the highest accuracy among the evaluated models (0.9802), but its lower LMOS value (0.7118) limits its ETI performance (0.8460). The results further demonstrate that models with slightly lower accuracy can outperform highly accurate models when they generate more meaningful visual explanations. For example, Xception achieves lower accuracy than DenseNet121 but obtains nearly the highest ETI due to superior localization behavior.

For the flower classification datasets, the ETI analysis reveals similar but dataset-dependent behavior. On the FLRS dataset, Xception achieves the highest ETI score of 0.7849, followed by DenseNet121 (0.7617) and MobileNetV2 (0.7311). Although DenseNet121 provides the highest accuracy (0.9046), Xception has achieved better overall ETI due to its higher LMOS value (0.6869). The FR also demonstrates the effectiveness of ETI in distinguishing models with similar predictive performance. Xception

achieves the highest ETI score (0.7531), while DenseNet121 obtains the highest accuracy (0.9124). However, the lower LMOS of DenseNet121 (0.5382) reduces its ETI to 0.7253. These results confirm that the proposed metric provides additional insights beyond conventional accuracy ranking.

The ablation results shown in Figure 6 further validate the contribution of both components of ETI. The “ETI without accuracy” variant, which considers only LMOS information, produces substantially lower values across all datasets because explanation quality alone cannot represent complete model reliability. Similarly, the “ETI without LMOS” variant, which depends only on classification accuracy, fails to capture whether the model focuses on meaningful regions. For instance, in the EPLI dataset, the Custom CNN achieves an accuracy-based ETI value of 0.4972 when LMOS is removed, which incorrectly suggests competitive trustworthiness compared with stronger architectures. However, the complete ETI value decreases to 0.5922 due to its poor

**Table 7**

Accuracy stability of different CNN architectures on EPLI under various input perturbations.

| Model             | Noise  | Blur   | Brightness | Hue    | Contrast | Gamma  | Motion Blur | Compression | Occlusion | Mean   |
|-------------------|--------|--------|------------|--------|----------|--------|-------------|-------------|-----------|--------|
| Custom CNN        | 0.9801 | 0.9958 | 0.9021     | 0.8711 | 0.8494   | 0.8033 | 0.9916      | 0.9986      | 0.9801    | 0.9302 |
| DenseNet121       | 0.9944 | 0.9845 | 1.0000     | 0.9986 | 0.9860   | 0.9916 | 0.9159      | 0.9930      | 1.0000    | 0.9849 |
| MobileNetV2       | 0.9441 | 0.9534 | 1.0000     | 0.9916 | 0.9902   | 0.9986 | 0.8816      | 0.9916      | 0.9830    | 0.9704 |
| InceptionResNetV2 | 0.9727 | 0.9829 | 0.9986     | 0.9986 | 0.9944   | 0.9915 | 0.9149      | 0.9887      | 0.9469    | 0.9766 |
| InceptionV3       | 0.9593 | 0.9771 | 0.9958     | 0.9972 | 0.9944   | 0.9915 | 0.9217      | 0.9757      | 0.9201    | 0.9703 |
| ResNet50          | 0.9816 | 0.6962 | 0.9955     | 0.9862 | 0.4599   | 0.8978 | 0.5316      | 0.9769      | 0.9113    | 0.8263 |
| VGG16             | 0.9828 | 0.9695 | 0.9958     | 0.9813 | 0.8380   | 0.9843 | 0.8284      | 0.9943      | 0.9481    | 0.9469 |
| Xception          | 0.9801 | 0.9902 | 0.9986     | 1.0000 | 0.9972   | 0.9958 | 0.9549      | 0.9958      | 0.9859    | 0.9887 |

**Table 8**

LMOS stability of different CNN architectures on EPLI under various input perturbations.

| Model             | Noise  | Blur   | Brightness | Hue    | Contrast | Gamma  | Motion Blur | Compression | Occlusion | Mean   |
|-------------------|--------|--------|------------|--------|----------|--------|-------------|-------------|-----------|--------|
| Custom CNN        | 0.7525 | 0.7421 | 0.5343     | 0.5701 | 0.4545   | 0.5053 | 0.7242      | 0.9071      | 0.7376    | 0.6586 |
| DenseNet121       | 0.8322 | 0.8151 | 0.9352     | 0.8723 | 0.8256   | 0.8398 | 0.6551      | 0.8103      | 0.8402    | 0.8251 |
| MobileNetV2       | 0.8099 | 0.8200 | 0.9303     | 0.8910 | 0.8382   | 0.8763 | 0.7212      | 0.8393      | 0.8373    | 0.8404 |
| InceptionResNetV2 | 0.6846 | 0.6978 | 0.8646     | 0.8556 | 0.7867   | 0.7693 | 0.5969      | 0.7433      | 0.7177    | 0.7463 |
| InceptionV3       | 0.6550 | 0.6384 | 0.8425     | 0.8184 | 0.7378   | 0.7450 | 0.5560      | 0.6294      | 0.6426    | 0.6961 |
| ResNet50          | 0.8417 | 0.7607 | 0.9332     | 0.9070 | 0.7830   | 0.8010 | 0.7666      | 0.8410      | 0.8187    | 0.8281 |
| VGG16             | 0.7576 | 0.7014 | 0.8812     | 0.7836 | 0.5362   | 0.7432 | 0.5007      | 0.7693      | 0.6564    | 0.7033 |
| Xception          | 0.8540 | 0.8611 | 0.9482     | 0.9489 | 0.9122   | 0.8802 | 0.7613      | 0.8712      | 0.8877    | 0.8805 |

localization capability. This demonstrates that LMOS provides essential complementary information that cannot be obtained from accuracy alone.

Across all datasets, the ablation trends consistently show that the complete ETI formulation produces a more balanced and discriminative evaluation compared with either individual component. The integration of accuracy and LMOS prevents the possibility of overestimation of models that achieve high classification scores through potentially unreliable visual reasoning. Overall, the experimental results from Table 6 demonstrate that ETI provides a more comprehensive measurement of CNNs by simultaneously evaluating predictive reliability and explanation quality.

### 6.3. Explainable Robustness Index (ERI)

#### EPLI

The ERI values of the evaluated CNN architectures have been assessed from both prediction stability and explanation stability. Table 7 shows that most architectures have maintained high accuracy stability under common perturbations, particularly brightness, hue, gamma, and compression. Xception has achieved the highest mean accuracy stability (0.9887), followed by DenseNet121 (0.9849) and InceptionResNetV2 (0.9766). In contrast, ResNet50 has exhibited substantially lower stability (0.8263), mainly due to its sensitivity to blur, contrast, and motion blur. Although the Custom CNN has maintained a relatively high mean accuracy stability of 0.9302, its performance does not necessarily indicate reliable robustness when the stability of its explanations is considered.

Table 8 reveals greater variation in explanation stability across architectures. Xception has achieved the highest mean LMOS stability (0.8805), followed by MobileNetV2 (0.8404) and ResNet50 (0.8281). The Custom CNN has shown considerably lower LMOS stability (0.6586), indicating that its Grad-CAM explanations have changed more

substantially under perturbations despite relatively stable predictions. Motion blur generally has produced the largest degradation in explanation stability, while brightness and hue shifts are comparatively less disruptive for most architectures.

The combined results are reflected in the ERI values in Table 9. Xception has achieved the highest overall ERI of 0.9346, demonstrating a strong balance between prediction and explanation stability across all perturbations. MobileNetV2 (0.9054) and DenseNet121 (0.9050) have followed closely. At the other end of the ranking, the Custom CNN has obtained the lowest ERI (0.7944), despite achieving a substantially higher accuracy stability than ResNet50. This demonstrates the importance of incorporating explanation stability into robustness assessment: high predictive stability alone does not guarantee stable and trustworthy model behavior. Across perturbations, brightness and hue generally have produced the highest ERI values, whereas motion blur is consistently among the most challenging perturbations.

The ranking and ablation results in Table 10 further support this observation. Xception has ranked first with an ERI of 0.9346, while MobileNetV2 and DenseNet121 have ranked second and third, respectively. Removing either accuracy stability or LMOS stability substantially reduces the resulting score for every architecture, confirming that both components contribute meaningfully to the proposed ERI. Overall, the EPLI results demonstrate that ERI provides a more comprehensive characterization of CNN robustness by jointly considering whether predictions remain stable and whether the corresponding visual explanations remain consistent under input perturbations.

#### BDML

The results on BDML further demonstrate the complementary roles of prediction and explanation stability in evaluating CNN robustness. As shown in Table 11, DenseNet121

**Table 9**

ERI of different CNN architectures on EPLI under individual input perturbations.

| Model             | Noise  | Blur   | Brightness | Hue    | Contrast | Gamma  | Motion Blur | Compression | Occlusion | Overall ERI |
|-------------------|--------|--------|------------|--------|----------|--------|-------------|-------------|-----------|-------------|
| Custom CNN        | 0.8663 | 0.8690 | 0.7182     | 0.7206 | 0.6520   | 0.6543 | 0.8579      | 0.9528      | 0.8589    | 0.7944      |
| DenseNet121       | 0.9133 | 0.8998 | 0.9676     | 0.9355 | 0.9058   | 0.9157 | 0.7855      | 0.9017      | 0.9201    | 0.9050      |
| MobileNetV2       | 0.8770 | 0.8867 | 0.9651     | 0.9413 | 0.9142   | 0.9375 | 0.8014      | 0.9155      | 0.9102    | 0.9054      |
| InceptionResNetV2 | 0.8286 | 0.8403 | 0.9316     | 0.9271 | 0.8905   | 0.8804 | 0.7559      | 0.8660      | 0.8323    | 0.8614      |
| InceptionV3       | 0.8072 | 0.8077 | 0.9192     | 0.9078 | 0.8661   | 0.8682 | 0.7389      | 0.8025      | 0.7813    | 0.8332      |
| ResNet50          | 0.9117 | 0.7285 | 0.9644     | 0.9466 | 0.6215   | 0.8494 | 0.6491      | 0.9089      | 0.8650    | 0.8272      |
| VGG16             | 0.8702 | 0.8355 | 0.9385     | 0.8824 | 0.6871   | 0.8638 | 0.6645      | 0.8818      | 0.8023    | 0.8251      |
| Xception          | 0.9170 | 0.9256 | 0.9734     | 0.9745 | 0.9547   | 0.9380 | 0.8581      | 0.9335      | 0.9368    | 0.9346      |

**Table 10**

ERI ranking and ablation analysis of CNN architectures on EPLI

| Rank | Model             | Accuracy Stability (mean) | LMOS Stability (mean) | ERI    | ERI w/o Accuracy | ERI w/o LMOS |
|------|-------------------|---------------------------|-----------------------|--------|------------------|--------------|
| 1    | Xception          | 0.9887                    | 0.8805                | 0.9346 | 0.4403           | 0.4944       |
| 2    | MobileNetV2       | 0.9704                    | 0.8404                | 0.9054 | 0.4202           | 0.4852       |
| 3    | DenseNet121       | 0.9849                    | 0.8251                | 0.9050 | 0.4125           | 0.4924       |
| 4    | InceptionResNetV2 | 0.9766                    | 0.7463                | 0.8614 | 0.3731           | 0.4883       |
| 5    | InceptionV3       | 0.9703                    | 0.6961                | 0.8332 | 0.3481           | 0.4852       |
| 6    | ResNet50          | 0.8263                    | 0.8281                | 0.8272 | 0.4140           | 0.4132       |
| 7    | VGG16             | 0.9469                    | 0.7033                | 0.8251 | 0.3517           | 0.4735       |
| 8    | Custom CNN        | 0.9302                    | 0.6586                | 0.7944 | 0.3293           | 0.4651       |

has achieved the highest mean accuracy stability (0.9665), followed by InceptionResNetV2 (0.9552), VGG16 (0.9526), MobileNetV2 (0.9523), and InceptionV3 (0.9544). Xception has achieved a slightly lower mean accuracy stability of 0.9471 but maintained consistently high stability across most perturbations. In contrast, ResNet50 (0.7553) and the Custom CNN (0.7864) have been substantially more sensitive to several perturbations, particularly blur, contrast, and motion blur.

The LMOS stability results in Table 12 show a different pattern. ResNet50 has achieved the highest mean LMOS stability (0.8565), closely followed by Xception (0.8493), whereas DenseNet121 and MobileNetV2 obtained 0.8130 and 0.7840, respectively. The Custom CNN exhibited

the lowest LMOS stability (0.5556), indicating substantial changes in its Grad-CAM localization under perturbations. Similar to EgyPLI, motion blur generally caused strong degradation in explanation stability, while brightness and hue were comparatively less disruptive for the better-performing architectures.

The resulting ERI values in Table 13 provide a more balanced assessment by jointly considering both dimensions. Xception has achieved the highest overall ERI (0.8982), followed by DenseNet121 (0.8897) and MobileNetV2 (0.8682). Although ResNet50 has exhibited the highest LMOS stability, its low accuracy stability has substantially reduced its overall ERI to 0.8059. Conversely, the Custom CNN has obtained the lowest ERI (0.6710), reflecting instability in both predictions and explanations. Across the perturbations,

**Table 11**

Accuracy stability of different CNN architectures on BDML under various input perturbations.

| Model             | Noise  | Blur   | Brightness | Hue    | Contrast | Gamma  | Motion Blur | Compression | Occlusion | Mean   |
|-------------------|--------|--------|------------|--------|----------|--------|-------------|-------------|-----------|--------|
| Custom CNN        | 0.9689 | 0.7265 | 0.7319     | 0.8359 | 0.5174   | 0.8221 | 0.5943      | 0.9724      | 0.9084    | 0.7864 |
| DenseNet121       | 0.9794 | 0.9496 | 0.9924     | 0.9846 | 0.9846   | 0.9898 | 0.8750      | 0.9714      | 0.9714    | 0.9665 |
| MobileNetV2       | 0.9792 | 0.9111 | 0.9949     | 1.0000 | 0.9871   | 0.9949 | 0.8085      | 0.9574      | 0.9377    | 0.9523 |
| InceptionResNetV2 | 0.9513 | 0.9360 | 0.9917     | 0.9917 | 0.9890   | 0.9973 | 0.8738      | 0.9204      | 0.9452    | 0.9552 |
| InceptionV3       | 0.9810 | 0.9379 | 0.9838     | 0.9920 | 0.9865   | 0.9973 | 0.8876      | 0.9195      | 0.9038    | 0.9544 |
| ResNet50          | 0.9780 | 0.4103 | 0.8929     | 0.9738 | 0.3966   | 0.8929 | 0.4103      | 0.9186      | 0.9249    | 0.7553 |
| VGG16             | 1.0000 | 0.9538 | 0.9916     | 0.9803 | 0.8997   | 0.9745 | 0.8131      | 0.9888      | 0.9716    | 0.9526 |
| Xception          | 0.9760 | 0.8739 | 0.9895     | 0.9921 | 0.9868   | 0.9948 | 0.8805      | 0.9122      | 0.9183    | 0.9471 |

**Table 12**

LMOS stability of different CNN architectures on BDML under various input perturbations.

| Model             | Noise  | Blur   | Brightness | Hue    | Contrast | Gamma  | Motion Blur | Compression | Occlusion | Mean   |
|-------------------|--------|--------|------------|--------|----------|--------|-------------|-------------|-----------|--------|
| Custom CNN        | 0.6527 | 0.5391 | 0.5192     | 0.5206 | 0.4322   | 0.4802 | 0.5138      | 0.7388      | 0.6041    | 0.5556 |
| DenseNet121       | 0.8505 | 0.7066 | 0.8788     | 0.8860 | 0.8255   | 0.8573 | 0.6538      | 0.8219      | 0.8365    | 0.8130 |
| MobileNetV2       | 0.8044 | 0.7037 | 0.8849     | 0.8624 | 0.7967   | 0.8231 | 0.6132      | 0.7769      | 0.7908    | 0.7840 |
| InceptionResNetV2 | 0.6655 | 0.6017 | 0.8148     | 0.7627 | 0.7656   | 0.7649 | 0.5920      | 0.6376      | 0.6674    | 0.6969 |
| InceptionV3       | 0.6297 | 0.5752 | 0.7254     | 0.7291 | 0.6905   | 0.6274 | 0.5267      | 0.5578      | 0.5659    | 0.6253 |
| ResNet50          | 0.9070 | 0.8051 | 0.8610     | 0.9044 | 0.7997   | 0.8418 | 0.8091      | 0.8880      | 0.8927    | 0.8565 |
| VGG16             | 0.8378 | 0.6823 | 0.7996     | 0.7550 | 0.6200   | 0.7023 | 0.4370      | 0.7505      | 0.7047    | 0.6988 |
| Xception          | 0.8693 | 0.7526 | 0.9185     | 0.9105 | 0.9097   | 0.8719 | 0.7333      | 0.8027      | 0.8747    | 0.8493 |

**Table 13**

ERI of different CNN architectures on BDML under individual input perturbations

| Model             | Noise  | Blur   | Brightness | Hue    | Contrast | Gamma  | Motion Blur | Compression | Occlusion | Overall ERI |
|-------------------|--------|--------|------------|--------|----------|--------|-------------|-------------|-----------|-------------|
| Custom CNN        | 0.8108 | 0.6328 | 0.6256     | 0.6783 | 0.4748   | 0.6512 | 0.5541      | 0.8556      | 0.7563    | 0.6710      |
| DenseNet121       | 0.9150 | 0.8281 | 0.9356     | 0.9353 | 0.9051   | 0.9236 | 0.7644      | 0.8967      | 0.9040    | 0.8897      |
| MobileNetV2       | 0.8918 | 0.8074 | 0.9399     | 0.9312 | 0.8919   | 0.9090 | 0.7109      | 0.8672      | 0.8643    | 0.8682      |
| InceptionResNetV2 | 0.8084 | 0.7689 | 0.9033     | 0.8772 | 0.8773   | 0.8811 | 0.7329      | 0.7790      | 0.8063    | 0.8260      |
| InceptionV3       | 0.8054 | 0.7566 | 0.8546     | 0.8606 | 0.8385   | 0.8124 | 0.7072      | 0.7387      | 0.7349    | 0.7898      |
| ResNet50          | 0.9425 | 0.6077 | 0.8769     | 0.9391 | 0.5982   | 0.8674 | 0.6097      | 0.9033      | 0.9088    | 0.8059      |
| VGG16             | 0.9189 | 0.8181 | 0.8956     | 0.8677 | 0.7599   | 0.8384 | 0.6251      | 0.8697      | 0.8382    | 0.8257      |
| Xception          | 0.9227 | 0.8133 | 0.9540     | 0.9513 | 0.9483   | 0.9334 | 0.8069      | 0.8575      | 0.8965    | 0.8982      |

**Table 14**

ERI ranking and ablation analysis of CNN architectures on BDML

| Rank | Model             | Accuracy Stability (mean) | LMOS Stability (mean) | ERI    | ERI w/o Accuracy | ERI w/o LMOS |
|------|-------------------|---------------------------|-----------------------|--------|------------------|--------------|
| 1    | Xception          | 0.9471                    | 0.8493                | 0.8982 | 0.4246           | 0.4736       |
| 2    | DenseNet121       | 0.9665                    | 0.8130                | 0.8897 | 0.4065           | 0.4832       |
| 3    | MobileNetV2       | 0.9523                    | 0.7840                | 0.8682 | 0.3920           | 0.4762       |
| 4    | InceptionResNetV2 | 0.9552                    | 0.6969                | 0.8260 | 0.3485           | 0.4776       |
| 5    | VGG16             | 0.9526                    | 0.6988                | 0.8257 | 0.3494           | 0.4763       |
| 6    | ResNet50          | 0.7553                    | 0.8565                | 0.8059 | 0.4283           | 0.3777       |
| 7    | InceptionV3       | 0.9544                    | 0.6253                | 0.7898 | 0.3127           | 0.4772       |
| 8    | Custom CNN        | 0.7864                    | 0.5556                | 0.6710 | 0.2778           | 0.3932       |

brightness and hue generally resulted in the highest ERI values, whereas contrast and motion blur have been among the most challenging transformations.

The final ranking and ablation analysis in Table 14 confirms these observations. Xception has ranked first, demonstrating the strongest overall balance between predictive and explanatory robustness. DenseNet121 and MobileNetV2 have followed in second and third positions. Notably, removing either accuracy stability or LMOS stability substantially reduced the resulting score for all architectures, again indicating that both components are necessary for a meaningful ERI assessment. Overall, the BDML results reinforce the ability of ERI to distinguish models that may appear robust when evaluated solely on prediction stability but exhibit substantially different behavior in their visual explanations.

#### FLRS

The results on the FLRS dataset further demonstrate the complementary roles of prediction and explanation stability in evaluating ERI. As shown in Table 15, Xception has achieved the highest mean accuracy stability (0.9876), closely followed by InceptionV3 (0.9860) and InceptionResNetV2 (0.9783). Most pretrained architectures have maintained high prediction stability across perturbations, whereas ResNet50 has exhibited comparatively lower stability (0.9064), particularly under contrast, blur, and motion blur. The Custom CNN has also shown reduced stability under contrast and hue perturbations despite maintaining a relatively high overall mean of 0.9528.

The LMOS stability results in Table 16 reveal greater variation across architectures. Xception has achieved the highest mean LMOS stability (0.8741), followed by ResNet50 (0.8347) and DenseNet121 (0.8286). In contrast, the Custom CNN and InceptionV3 have obtained substantially lower values of 0.6763 and 0.7326, respectively. Brightness and

hue perturbations generally have preserved explanation stability, whereas contrast and motion blur have produced larger degradations for several models. These results indicate that models with stable predictions may still exhibit considerable changes in their explanatory regions.

Combining the two stability components, Table 17 shows that Xception has achieved the highest overall ERI (0.9308), followed by DenseNet121 (0.8997), VGG16 (0.8968), and InceptionResNetV2 (0.8900). Xception has consistently maintained high ERI across individual perturbations, with particularly strong performance under brightness (0.9557), hue (0.9550), and contrast (0.9457). In contrast, the Custom CNN has obtained the lowest overall ERI (0.8146), mainly due to its lower LMOS stability. Contrast and motion blur are among the more challenging perturbations for several architectures, while brightness and hue are generally less disruptive.

The ranking and ablation analysis in Table 18 further confirms the importance of incorporating both components. Xception has ranked first with an ERI of 0.9308, followed by DenseNet121 (0.8997) and VGG16 (0.8968). Although InceptionV3 has achieved very high accuracy stability (0.9860), its lower LMOS stability (0.7326) reduced its overall ERI to 0.8593. Similarly, ResNet50 has achieved the second-highest LMOS stability (0.8347), but its lower accuracy stability (0.9064) compared to others limited its ERI to 0.8706. Removing either accuracy or LMOS stability substantially reduces the resulting score, demonstrating that ERI provides a more balanced assessment of CNNs' robustness than either predictive or explanatory stability alone.

#### FR

The results on the FR further validate the effectiveness of ERI in jointly evaluating predictive and explanatory robustness. As shown in Table 19, Xception has achieved the highest mean accuracy stability (0.9901), followed by

**Table 15**

Accuracy stability of different CNN architectures on the FLRS dataset under various input perturbations.

| Model             | Noise  | Blur   | Brightness | Hue    | Contrast | Gamma  | Motion Blur | Compression | Occlusion | Mean   |
|-------------------|--------|--------|------------|--------|----------|--------|-------------|-------------|-----------|--------|
| Custom CNN        | 0.9902 | 0.9842 | 0.9553     | 0.8850 | 0.8527   | 0.9553 | 0.9741      | 0.9961      | 0.9822    | 0.9528 |
| DenseNet121       | 0.9847 | 0.9510 | 0.9909     | 0.9657 | 0.9878   | 0.9785 | 0.9273      | 0.9816      | 0.9689    | 0.9707 |
| MobileNetV2       | 0.9808 | 0.9539 | 0.9921     | 0.9921 | 0.9725   | 0.9905 | 0.9435      | 0.9742      | 0.9675    | 0.9741 |
| InceptionResNetV2 | 0.9770 | 0.9703 | 0.9870     | 0.9887 | 0.9984   | 0.9935 | 0.9720      | 0.9804      | 0.9370    | 0.9783 |
| InceptionV3       | 0.9967 | 0.9849 | 0.9950     | 0.9933 | 0.9917   | 0.9967 | 0.9477      | 0.9797      | 0.9883    | 0.9860 |
| ResNet50          | 1.0000 | 0.8287 | 0.9556     | 0.9920 | 0.6714   | 0.9783 | 0.7871      | 0.9613      | 0.9838    | 0.9064 |
| VGG16             | 0.9983 | 0.9725 | 0.9950     | 0.9672 | 0.9167   | 0.9966 | 0.9283      | 0.9967      | 0.9795    | 0.9723 |
| Xception          | 0.9985 | 0.9843 | 0.9938     | 1.0000 | 0.9922   | 0.9985 | 0.9682      | 0.9763      | 0.9763    | 0.9876 |

**Table 16**

LMOS stability of different CNN architectures on the FLRS dataset under various input perturbations.

| Model             | Noise  | Blur   | Brightness | Hue    | Contrast | Gamma  | Motion Blur | Compression | Occlusion | Mean   |
|-------------------|--------|--------|------------|--------|----------|--------|-------------|-------------|-----------|--------|
| Custom CNN        | 0.7787 | 0.7326 | 0.6323     | 0.5899 | 0.5376   | 0.6056 | 0.6934      | 0.8218      | 0.6951    | 0.6763 |
| DenseNet121       | 0.8577 | 0.7766 | 0.8921     | 0.8085 | 0.8384   | 0.8277 | 0.7809      | 0.8376      | 0.8382    | 0.8286 |
| MobileNetV2       | 0.7818 | 0.7627 | 0.8826     | 0.8121 | 0.7751   | 0.8115 | 0.7459      | 0.7510      | 0.8251    | 0.7942 |
| InceptionResNetV2 | 0.8068 | 0.7550 | 0.8578     | 0.8547 | 0.8234   | 0.8297 | 0.7444      | 0.7844      | 0.7600    | 0.8018 |
| InceptionV3       | 0.7358 | 0.6701 | 0.8221     | 0.7992 | 0.7877   | 0.7356 | 0.6244      | 0.6879      | 0.7307    | 0.7326 |
| ResNet50          | 0.9087 | 0.7728 | 0.8949     | 0.8864 | 0.7239   | 0.8407 | 0.7469      | 0.8621      | 0.8762    | 0.8347 |
| VGG16             | 0.8757 | 0.8135 | 0.8687     | 0.8143 | 0.7450   | 0.8218 | 0.7380      | 0.8750      | 0.8403    | 0.8214 |
| Xception          | 0.8573 | 0.8562 | 0.9176     | 0.9100 | 0.8993   | 0.8870 | 0.8249      | 0.8526      | 0.8619    | 0.8741 |

InceptionResNetV2 (0.9811), DenseNet121 (0.9808), and InceptionV3 (0.9795). Most pretrained architectures have maintained high prediction stability across perturbations, whereas ResNet50 has shown substantially lower stability (0.8944), particularly under contrast and motion blur. The Custom CNN has also exhibited reduced stability under hue and contrast perturbations.

The LMOS stability results in Table 20 show a clearer distinction among the models. Xception has achieved the highest mean LMOS stability (0.9478), substantially exceeding the other architectures. DenseNet121 and ResNet50 followed with 0.8978 and 0.8960, respectively, while the Custom CNN and InceptionV3 obtained the lowest values

of 0.7798 and 0.7858. Xception has also maintained consistently high explanation stability across all perturbations, including brightness (0.9591), hue (0.9667), contrast (0.9607), and gamma (0.9653), indicating that its Grad-CAM localization was comparatively insensitive to input changes.

By combining both stability components, Table 21 shows that Xception has achieved the highest overall ERI (0.9689), followed by DenseNet121 (0.9393), InceptionResNetV2 (0.9273), and VGG16 (0.9262). Xception has consistently achieved the highest ERI across all individual perturbations, with values above 0.94 for every perturbation and particularly strong performance under brightness (0.9789), hue (0.9821), contrast (0.9791), and gamma (0.9775). In contrast, the Custom CNN has obtained the lowest overall ERI (0.8680), primarily due to its weaker LMOS stability.

**Table 17**

ERI of different CNN architectures on the FLRS dataset under individual input perturbations.

| Model             | Noise  | Blur   | Brightness | Hue    | Contrast | Gamma  | Motion Blur | Compression | Occlusion | Overall ERI |
|-------------------|--------|--------|------------|--------|----------|--------|-------------|-------------|-----------|-------------|
| Custom CNN        | 0.8844 | 0.8584 | 0.7938     | 0.7374 | 0.6951   | 0.7805 | 0.8337      | 0.9089      | 0.8387    | 0.8146      |
| DenseNet121       | 0.9212 | 0.8638 | 0.9415     | 0.8871 | 0.9131   | 0.9031 | 0.8541      | 0.9096      | 0.9035    | 0.8997      |
| MobileNetV2       | 0.8813 | 0.8583 | 0.9374     | 0.9021 | 0.8738   | 0.9010 | 0.8447      | 0.8626      | 0.8963    | 0.8842      |
| InceptionResNetV2 | 0.8919 | 0.8627 | 0.9224     | 0.9217 | 0.9109   | 0.9116 | 0.8582      | 0.8824      | 0.8485    | 0.8900      |
| InceptionV3       | 0.8662 | 0.8275 | 0.9085     | 0.8962 | 0.8897   | 0.8661 | 0.7860      | 0.8338      | 0.8595    | 0.8593      |
| ResNet50          | 0.9544 | 0.8008 | 0.9253     | 0.9392 | 0.6976   | 0.9095 | 0.7670      | 0.9117      | 0.9300    | 0.8706      |
| VGG16             | 0.9370 | 0.8930 | 0.9319     | 0.8907 | 0.8308   | 0.9092 | 0.8332      | 0.9359      | 0.9099    | 0.8968      |
| Xception          | 0.9279 | 0.9203 | 0.9557     | 0.9550 | 0.9457   | 0.9427 | 0.8965      | 0.9144      | 0.9191    | 0.9308      |

**Table 18**

ERI ranking and ablation analysis of CNN architectures on the FLRS dataset

| Rank | Model             | Accuracy Stability (mean) | LMOS Stability (mean) | ERI    | ERI w/o Accuracy | ERI w/o LMOS |
|------|-------------------|---------------------------|-----------------------|--------|------------------|--------------|
| 1    | Xception          | 0.9876                    | 0.8741                | 0.9308 | 0.4371           | 0.4938       |
| 2    | DenseNet121       | 0.9707                    | 0.8286                | 0.8997 | 0.4143           | 0.4854       |
| 3    | VGG16             | 0.9723                    | 0.8214                | 0.8968 | 0.4107           | 0.4862       |
| 4    | InceptionResNetV2 | 0.9783                    | 0.8018                | 0.8900 | 0.4009           | 0.4891       |
| 5    | MobileNetV2       | 0.9741                    | 0.7942                | 0.8842 | 0.3971           | 0.4871       |
| 6    | ResNet50          | 0.9064                    | 0.8347                | 0.8706 | 0.4174           | 0.4532       |
| 7    | InceptionV3       | 0.9860                    | 0.7326                | 0.8593 | 0.3663           | 0.4930       |
| 8    | Custom CNN        | 0.9528                    | 0.6763                | 0.8146 | 0.3382           | 0.4764       |

**Table 19**

Accuracy stability of different CNN architectures on the FR under various input perturbations.

| Model             | Noise  | Blur   | Brightness | Hue    | Contrast | Gamma  | Motion Blur | Compression | Occlusion | Mean   |
|-------------------|--------|--------|------------|--------|----------|--------|-------------|-------------|-----------|--------|
| Custom CNN        | 0.9933 | 0.9828 | 0.9504     | 0.8872 | 0.8851   | 0.9704 | 0.9811      | 0.9949      | 0.9614    | 0.9563 |
| DenseNet121       | 0.9975 | 0.9767 | 0.9987     | 0.9781 | 0.9781   | 0.9898 | 0.9468      | 0.9741      | 0.9872    | 0.9808 |
| MobileNetV2       | 0.9750 | 0.9402 | 0.9896     | 0.9883 | 0.9857   | 0.9935 | 0.9516      | 0.9791      | 0.9791    | 0.9758 |
| InceptionResNetV2 | 0.9788 | 0.9802 | 0.9936     | 1.0000 | 0.9909   | 0.9909 | 0.9734      | 0.9869      | 0.9352    | 0.9811 |
| InceptionV3       | 0.9841 | 0.9773 | 0.9961     | 0.9961 | 0.9841   | 0.9908 | 0.9521      | 0.9800      | 0.9550    | 0.9795 |
| ResNet50          | 0.9640 | 0.8449 | 0.9714     | 0.9835 | 0.6584   | 0.9739 | 0.7514      | 0.9307      | 0.9714    | 0.8944 |
| VGG16             | 0.9931 | 0.9555 | 0.9945     | 0.9761 | 0.9102   | 0.9832 | 0.8851      | 0.9986      | 0.9832    | 0.9644 |
| Xception          | 0.9910 | 0.9962 | 0.9987     | 0.9975 | 0.9975   | 0.9897 | 0.9752      | 0.9831      | 0.9818    | 0.9901 |

**Table 20**

LMOS stability of different CNN architectures on the FR under various input perturbations.

| Model             | Noise  | Blur   | Brightness | Hue    | Contrast | Gamma  | Motion Blur | Compression | Occlusion | Mean   |
|-------------------|--------|--------|------------|--------|----------|--------|-------------|-------------|-----------|--------|
| Custom CNN        | 0.8703 | 0.8340 | 0.7432     | 0.7218 | 0.6697   | 0.7233 | 0.8263      | 0.8870      | 0.7425    | 0.7798 |
| DenseNet121       | 0.9124 | 0.8823 | 0.9376     | 0.8907 | 0.9138   | 0.9102 | 0.8383      | 0.9085      | 0.8867    | 0.8978 |
| MobileNetV2       | 0.8420 | 0.8128 | 0.9088     | 0.8817 | 0.8623   | 0.8572 | 0.8231      | 0.8321      | 0.8666    | 0.8541 |
| InceptionResNetV2 | 0.8762 | 0.8427 | 0.9283     | 0.8963 | 0.9031   | 0.8905 | 0.8281      | 0.8689      | 0.8281    | 0.8736 |
| InceptionV3       | 0.7838 | 0.7453 | 0.8439     | 0.8416 | 0.8217   | 0.7972 | 0.7059      | 0.7697      | 0.7634    | 0.7858 |
| ResNet50          | 0.9363 | 0.8639 | 0.9226     | 0.9181 | 0.8441   | 0.9022 | 0.8536      | 0.9011      | 0.9223    | 0.8960 |
| VGG16             | 0.9321 | 0.8797 | 0.9242     | 0.8781 | 0.8303   | 0.9000 | 0.8449      | 0.9180      | 0.8846    | 0.8880 |
| Xception          | 0.9425 | 0.9469 | 0.9591     | 0.9667 | 0.9607   | 0.9653 | 0.9154      | 0.9339      | 0.9398    | 0.9478 |

Contrast and motion blur were among the more challenging perturbations, particularly for architectures with lower predictive or explanatory stability.

The ranking and ablation results in Table 22 further emphasize the complementary nature of the two ERI components. Xception has ranked first with an ERI of 0.9689, while DenseNet121 ranked second with 0.9393. Notably, ResNet50 has achieved a relatively high LMOS stability (0.8960) but had substantially lower accuracy stability (0.8944), resulting in an ERI of 0.8952. Conversely, InceptionV3 has maintained very high accuracy stability (0.9795) but had considerably lower LMOS stability (0.7858), yielding an ERI of only 0.8827. Thus, high predictive stability or high explanatory stability alone is insufficient to ensure a high ERI. The substantial reduction in score when either

component is excluded further supports the use of both prediction and explanation stability for a more comprehensive assessment of CNN reliability.

#### 6.4. Cross-Dataset Analysis of ETI and ERI

The cross-dataset results provide a broader assessment of the proposed ETI and ERI metrics and reveal several consistent patterns across different visual domains. As shown in Tables 6 and the corresponding ERI results, the relative performance of the CNN architectures is influenced not only by their classification capability but also by the quality and stability of their visual explanations. While ETI evaluates a model by jointly considering classification accuracy and mean LMOS, ERI focuses on the stability of both prediction and explanation under input perturbations. Therefore, the

**Table 21**

ERI of different CNN architectures on the FR under individual input perturbations.

| Model             | Noise  | Blur   | Brightness | Hue    | Contrast | Gamma  | Motion Blur | Compression | Occlusion | Overall ERI |
|-------------------|--------|--------|------------|--------|----------|--------|-------------|-------------|-----------|-------------|
| Custom CNN        | 0.9318 | 0.9084 | 0.8468     | 0.8045 | 0.7774   | 0.8469 | 0.9037      | 0.9409      | 0.8519    | 0.8680      |
| DenseNet121       | 0.9549 | 0.9295 | 0.9682     | 0.9344 | 0.9459   | 0.9500 | 0.8925      | 0.9413      | 0.9369    | 0.9393      |
| MobileNetV2       | 0.9085 | 0.8765 | 0.9492     | 0.9350 | 0.9240   | 0.9254 | 0.8874      | 0.9056      | 0.9228    | 0.9149      |
| InceptionResNetV2 | 0.9275 | 0.9114 | 0.9609     | 0.9482 | 0.9470   | 0.9407 | 0.9007      | 0.9279      | 0.8817    | 0.9273      |
| InceptionV3       | 0.8840 | 0.8613 | 0.9200     | 0.9188 | 0.9029   | 0.8940 | 0.8290      | 0.8749      | 0.8592    | 0.8827      |
| ResNet50          | 0.9502 | 0.8544 | 0.9470     | 0.9508 | 0.7512   | 0.9381 | 0.8025      | 0.9159      | 0.9468    | 0.8952      |
| VGG16             | 0.9626 | 0.9176 | 0.9594     | 0.9271 | 0.8702   | 0.9416 | 0.8650      | 0.9583      | 0.9339    | 0.9262      |
| Xception          | 0.9667 | 0.9715 | 0.9789     | 0.9821 | 0.9791   | 0.9775 | 0.9453      | 0.9585      | 0.9608    | 0.9689      |

**Table 22**

ERI ranking and ablation analysis of CNN architectures on the FR

| Rank | Model             | Accuracy Stability (mean) | LMOS Stability (mean) | ERI    | ERI w/o Accuracy | ERI w/o LMOS |
|------|-------------------|---------------------------|-----------------------|--------|------------------|--------------|
| 1    | Xception          | 0.9901                    | 0.9478                | 0.9689 | 0.4739           | 0.4950       |
| 2    | DenseNet121       | 0.9808                    | 0.8978                | 0.9393 | 0.4489           | 0.4904       |
| 3    | InceptionResNetV2 | 0.9811                    | 0.8736                | 0.9273 | 0.4368           | 0.4905       |
| 4    | VGG16             | 0.9644                    | 0.8880                | 0.9262 | 0.4440           | 0.4822       |
| 5    | MobileNetV2       | 0.9758                    | 0.8541                | 0.9149 | 0.4270           | 0.4879       |
| 6    | ResNet50          | 0.8944                    | 0.8960                | 0.8952 | 0.4480           | 0.4472       |
| 7    | InceptionV3       | 0.9795                    | 0.7858                | 0.8827 | 0.3929           | 0.4898       |
| 8    | Custom CNN        | 0.9563                    | 0.7798                | 0.8680 | 0.3899           | 0.4781       |

two metrics provide complementary perspectives on CNN reliability.

Across the four datasets, ETI shows a clear distinction between the two leaf datasets and the two flower datasets. The highest ETI values are observed on the leaf datasets, EPLI and BDML. Here, MobileNetV2 has achieved 0.8855 and 0.8657, respectively, while Xception has achieved 0.8791 and 0.8653. In comparison, the highest ETI values on the FLRS and FR are 0.7849 and 0.7531, respectively, both achieved by Xception. This difference indicates that the models generally achieve stronger combinations of classification accuracy and localization quality on the leaf datasets than on the flower datasets. The lower ETI values on the flower datasets are mainly associated with reduced LMOS values, indicating that accurately recognizing flower categories does not necessarily imply that the models consistently focus on the corresponding flower regions.

Xception demonstrates particularly strong cross-dataset consistency for ETI. It ranks second on EPLI and BDML and first on both flower datasets. More importantly, Xception maintains relatively high LMOS values across all four datasets, ranging from 0.6029 on FR to 0.7800 on BDML. This consistency allows Xception to maintain a high ETI even when another architecture achieves slightly higher classification accuracy. For example, DenseNet121 has achieved the highest accuracy on EPLI, BDML, FLRS, and FR; especially on EPLI, it reaches 1.0000, yet it does not obtain the highest ETI on any of the four datasets. This provides strong cross-dataset evidence that classification accuracy alone is insufficient for assessing model trustworthiness. On EPLI, the Custom CNN has achieved 99.45 accuracy, but its object reliance is the lowest (only 0.19). ETI of the Custom CNN is always low, even when its accuracy is high.

The cross-dataset ERI results reveal a related but distinct pattern. Xception achieves the highest overall ERI on all four datasets, with values of 0.9346 on EgyPLI, 0.8982 on BDMediLeaves, 0.9308 on Flowers, and 0.9689 on FR. Its consistently strong performance in both accuracy stability and LMOS stability indicates that the architecture is comparatively resilient to input perturbations while preserving its explanation patterns. Particularly notable is its performance on the FR, where Xception achieves an accuracy stability of 0.9901 and an LMOS stability of 0.9478, resulting in an ERI of 0.9689. Furthermore, Xception remains the strongest model across individual perturbations on this dataset, demonstrating that its robustness is not attributable to a small number of favorable transformations.

An important observation from the cross-dataset comparison is that ETI and ERI do not necessarily produce identical performance patterns. The FLRS dataset, for example, has a substantially lower ETI for Xception (0.7849) than EPLI (0.8791), but its ERI is very high (0.9308). Similarly, the FR produces the highest ERI for Xception (0.9689) despite having a lower ETI (0.7531). This difference is expected because ETI incorporates the absolute classification accuracy and mean LMOS, whereas ERI evaluates how consistently these properties are preserved when the input

is perturbed. Consequently, a model may have moderate absolute performance but exhibit highly stable predictions and explanations, resulting in a high ERI. Thus, ETI and ERI should be interpreted as complementary rather than interchangeable measures.

The results also demonstrate that weak explanatory behavior can remain a major limitation even when classification performance is high. The Custom CNN provides a clear example across the datasets. On EgyPLI, it achieves an accuracy of 0.9945 but has a very low LMOS of 0.1900, resulting in the lowest ETI of 0.5922. A similar pattern is observed on BDMediLeaves, Flowers, and the FR, where the Custom CNN consistently obtains among the lowest LMOS and ETI values. Its ERI performance is also comparatively weak, particularly because its explanation stability remains lower than that of the strongest pretrained architectures. These findings suggest that the proposed metrics can identify models whose apparent predictive performance is not accompanied by reliable or stable visual reasoning.

ResNet50 provides another useful cross-dataset example of the complementary nature of the two metrics. It exhibits relatively high LMOS or LMOS stability in several datasets but suffers from substantially lower classification accuracy or accuracy stability. For instance, its accuracy on BDML, FLRS, and FR is only 0.4604, 0.5123, and 0.4977, respectively. Consequently, its ETI remains low despite comparatively reasonable localization scores. In the ERI evaluation, ResNet50 similarly demonstrates that strong explanation stability alone cannot compensate for poor predictive stability. This reinforces the central principle of the proposed framework: reliable CNN behavior requires both predictive and explanatory consistency.

Overall, the cross-dataset analysis demonstrates that ETI and ERI capture different but complementary aspects of CNN reliability. ETI is particularly effective for distinguishing models according to their joint classification performance and explanation quality, whereas ERI assesses whether both properties remain stable under realistic input perturbations. Xception consistently emerges as the most reliable architecture across the four datasets, achieving the highest ERI on every dataset and the highest ETI on both flower datasets. In contrast, models such as the Custom CNN can achieve high classification accuracy while exhibiting substantially weaker explanation quality and robustness. These consistent cross-dataset observations provide further evidence that evaluating CNNs through accuracy alone can overlook important aspects of model reliability, while the combined use of ETI and ERI offers a more comprehensive assessment of trustworthy and robust visual recognition.

Despite the strengths of this study, there are some limitations. ETI and ERI consider only two properties: accuracy and object focus alignment. To evaluate the whole scenario, many properties may need to be considered. Some research raises questions about Explainable AI methods such as Grad-CAM [39][40][41]. In the future, we have a plan to extend ETI and ERI to other XAI methods such as Grad-CAM++ [40], Score-CAM [42], LayerCAM [43], and Rise

[44] to assess the sensitivity of the proposed metrics to the choice of explainability technique. We will try to adapt ETI and ERI for other computer vision tasks, including object detection, semantic segmentation, and vision-language models.

## 6.5. Cross-Dataset Analysis

## 7. Related Works

### 7.1. Model Reliability and Trustworthiness

To move beyond standard classification accuracy, recent studies have increasingly focused on quantifying model reliability through uncertainty estimation and calibration. Javed et al. [14] integrated Bayesian Monte Carlo Dropout with nonparametric Conformal Prediction to address overconfidence in deep neural networks. To better evaluate model performance under uncertainty, Ao et al. [15] introduced the Excess Area Under the Optimal Risk-Coverage Curve (E-AUoptRC) and the Trust Index (TI), offering a risk-coverage perspective on model trustworthiness.

At a broader level, Nastoska et al. [45] provided a comprehensive review of AI trustworthiness criteria by evaluating industrial compliance against the NIST AI Risk Management Framework and ISO/IEC standards. In domain-specific applications, Pal et al. [46] proposed the Trust-Aware Explainable Artificial Intelligence (TAXAI) framework, which yields a normalized score by combining algorithmic transparency, interpretability alignment, and compliance reliability. While these frameworks advance the quantification of predictive trust, they generally evaluate reliability without explicitly measuring the spatial fidelity or stability of visual explanations under environmental perturbations.

### 7.2. Predictive Robustness Under Perturbations

Evaluating model performance under distribution shifts and input corruptions is essential for reliable deployment. Hendrycks and Dietterich [16] established foundational benchmarks known as ImageNet-C and ImageNet-P, to measure classification degradation under common corruptions and perturbations. Extending this, Schneider et al. [47] demonstrated that test-time covariate-shift adaptation can substantially recover predictive performance under corruptions, while Hendrycks et al. [17] proposed AugMix to improve model robustness while preserving clean accuracy. Beyond common corruptions, Su et al. [19] and Taori et al. [18] investigated adversarial and natural distribution shifts, demonstrating that high standard accuracy does not guarantee robustness and advocating for independent robustness evaluations. Furthermore, Guo et al. [20] compiled a 23-metric framework targeting adversarial data and model vulnerabilities.

While these works establish rigorous protocols for evaluating accuracy under stress, they focus exclusively on output predictions. A few recent efforts have attempted to incorporate explainability into robustness routines. Zhang et al. [48] proposed AR<sup>2</sup>, aligning Class Activation Maps (CAMs) between clean and corrupted images during training to enforce feature consistency. However, AR<sup>2</sup> operates as a model

repair mechanism rather than a quantitative evaluation index. In contrast, our proposed Explainable Robustness Index (ERI) directly quantifies the joint stability of both predictive performance and Grad-CAM visual evidence under input perturbations.

### 7.3. Explanation Quality, Localization, and Spurious Bias

Visual explanation techniques e.g., Grad-CAM, Attention maps, and Layer-wise Relevance Propagation (LRP) are widely deployed to interpret CNN decisions in sensitive domains, including smart farming and plant pathology [49, 50, 51]. However, as highlighted by Shawon et al. [51], high accuracy and visual heatmaps do not automatically guarantee model generalization under real-world variability. A critical challenge remains the reliance on shortcut learning and background bias. Bassi et al. [11] demonstrated that synthetic background biases heavily influence deep models, showing that LRP heatmaps can expose and mitigate shortcut learning without incurring high computational costs. Similarly, Zheng et al. [52] introduced ShortcutProbe to detect and retrain networks against prediction shortcuts without manual annotations.

A fundamental bottleneck in Explainable AI (XAI) is the scarcity of quantitative metrics for evaluating explanation quality. Addressing this, Rahimiaghdam and Alemdar [22] formulated evaluative metrics for medical imaging, while Baniecki et al. [53] proposed aligning model explanation pipelines directly with ground-truth segmentation masks. Ground-truth-based evaluation was further advanced by Guidotti et al. [24], who evaluated local explanation methods against annotated object bounds. Inspired by this perspective, our proposed Localization Mask Overlap Score (LMOS) quantifies the spatial alignment between Grad-CAM visual evidence and target object regions.

Regarding explanation stability, Repetto et al. [21] systematically measured the sensitivity of XAI techniques under natural noise and adversarial manipulations that alter explanations without affecting predictions. Finally, Fong and Vedaldi [23] and Kim et al. [54] analyzed black-box attribution and weakly supervised localization, observing that CAM-based methods often concentrate on localized, highly discriminative parts rather than full object extents. Our framework builds upon these insights by offering a unified, quantitative evaluation of spatial overlap (LMOS), explanation quality (ETI), and perturbation stability (ERI).

## 8. Conclusion

This paper has proposed two novel metrics, the Explainable Trustworthy Index (ETI) and the Explainable Robustness Index (ERI), to evaluate CNN models beyond classification accuracy. ETI combines accuracy with the Localization Mask Overlap Score (LMOS) to assess whether a model's predictions are grounded in the intended object rather than background or irrelevant regions. ERI extends this evaluation to perturbed inputs by jointly measuring

accuracy stability and LMOS stability, capturing whether both predictive performance and object-focused reasoning remain consistent under real-world degradations.

Across four benchmark datasets and eight CNN architectures, our results demonstrate that high classification accuracy does not guarantee high ETI or ERI. For example, on the EPLI dataset, the CustomCNN model has achieved competitive or near-perfect accuracy while obtaining the lowest ETI and ERI scores due to poor and unstable object localization and its poor stability. Similarly, the InceptionV3 achieves high classification accuracy across most conditions, its low ETI score reflects poor spatial focus on target object regions. Conversely, Xception consistently has achieved the highest overall ERI across all four datasets and the highest ETI on both flower datasets (FIRS and FR), despite not always achieving the highest raw accuracy. It illustrates that predictive performance and explanation quality/stability are related but distinct properties of model reliability. MobileNetV2 has outperformed higher-accuracy models on the BDML dataset in terms of ETI, reinforcing this distinction.

The cross-dataset comparison further shows that reliability patterns vary with visual domain: leaf datasets (EPLI and BDML) have generally yielded higher ETI than flower datasets (FLRS and FR), primarily due to lower LMOS on flowers, yet this does not translate into lower robustness. Several models have attained high ERI on the flower datasets despite modest ETI. This confirms that clean-image trustworthiness (ETI) and perturbation robustness (ERI) are complementary rather than interchangeable measures of reliability.

Overall, ETI and ERI provide a more comprehensive framework for evaluating CNNs in safety-critical or trust-sensitive applications, where accuracy alone is an insufficient proxy for whether a model reasons about the correct visual evidence and whether that reasoning holds up under realistic input perturbations.

## Data and Code availability

Supplementary code and data (including .json files and .keras files) associated with this research are available at:

[https://github.com/ragib1996/ETI\\_ERI](https://github.com/ragib1996/ETI_ERI)

## CRedit authorship contribution statement

**Ragib Mehedi:** Conceptualization, Methodology, Software, Investigation, Formal analysis, Writing – Original Draft, Writing – Review & Editing. **Azmain Inquaid Haque:** Investigation, Data Curation, Software, Validation. **Md. Nazmul Abdal:** Data Curation. **Abdullah Al Noman:** Supervision, Writing – Review & Editing. **Md. Zahidul Islam:** Supervision, Writing – Review & Editing.

## Acknowledgments

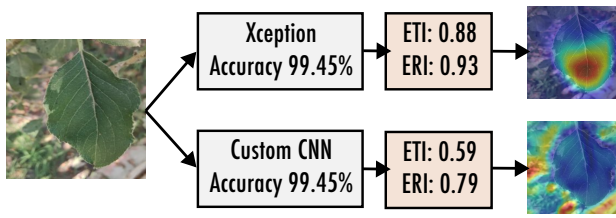
## Disclosure statement

## Funding

## References

- [1] R. Archana and P. E. Jeevaraj, “Deep learning models for digital image processing: a review,” *Artificial intelligence review*, vol. 57, no. 1, p. 11, 2024.
- [2] D. Shen, G. Wu, and H.-I. Suk, “Deep learning in medical image analysis,” *Annual review of biomedical engineering*, vol. 19, pp. 221–248, 2017.
- [3] A. Abade, P. A. Ferreira, and F. de Barros Vidal, “Plant diseases recognition on images using convolutional neural networks: A systematic review,” *Computers and Electronics in Agriculture*, vol. 185, p. 106125, 2021.
- [4] F. Lanusse *et al.*, “The dawes review 10: The impact of deep learning for the analysis of galaxy surveys,” *Publications of the Astronomical Society of Australia*, vol. 40, p. e001, 2023.
- [5] A. Zafar, L. K. Murali, and A. Vashist, “Beyond accuracy: Evaluating visual grounding in multimodal medical reasoning,” *arXiv preprint arXiv:2603.03437*, 2026.
- [6] R. R. Selvaraju, M. Cogswell, A. Das, R. Vedantam, D. Parikh, and D. Batra, “Grad-cam: Visual explanations from deep networks via gradient-based localization,” in *Proceedings of the IEEE international conference on computer vision*, pp. 618–626, 2017.
- [7] A. Nguyen, J. Yosinski, and J. Clune, “Deep neural networks are easily fooled: High confidence predictions for unrecognizable images,” in *Proceedings of the IEEE conference on computer vision and pattern recognition*, pp. 427–436, 2015.
- [8] R. Geirhos, P. Rubisch, C. Michaelis, M. Bethge, F. A. Wichmann, and W. Brendel, “Imagenet-trained cnns are biased towards texture; increasing shape bias improves accuracy and robustness,” in *International conference on learning representations*, 2018.
- [9] C. Szegedy, W. Zaremba, I. Sutskever, J. Bruna, D. Erhan, I. Goodfellow, and R. Fergus, “Intriguing properties of neural networks,” *arXiv preprint arXiv:1312.6199*, 2013.
- [10] J. Wu, W. Kang, H. Tang, Y. Hong, and Y. Yan, “On the faithfulness of vision transformer explanations,” in *2024 IEEE/CVF Conference on Computer Vision and Pattern Recognition (CVPR)*, pp. 10936–10945, IEEE, 2024.
- [11] P. R. Bassi, S. S. Dertkigil, and A. Cavalli, “Improving deep neural network generalization and robustness to background bias via layer-wise relevance propagation optimization,” *Nature Communications*, vol. 15, no. 1, p. 291, 2024.
- [12] R. Geirhos, J.-H. Jacobsen, C. Michaelis, R. Zemel, W. Brendel, M. Bethge, and F. A. Wichmann, “Shortcut learning in deep neural networks,” *Nature Machine Intelligence*, vol. 2, no. 11, pp. 665–673, 2020.
- [13] K. Xiao, L. Engstrom, A. Ilyas, and A. Madry, “Noise or signal: The role of image backgrounds in object recognition,” *arXiv preprint arXiv:2006.09994*, 2020.
- [14] F. Javed, S. Ruijs, and A. Kosiakova, “Beyond accuracy: Reliability and uncertainty estimation in convolutional neural networks,” *Available at SSRN 6030753*.
- [15] S. Ao, S. Rueger, and A. Siddharthan, “Empirical optimal risk to quantify model trustworthiness for failure detection,” *arXiv preprint arXiv:2308.03179*, 2023.
- [16] D. Hendrycks and T. Dietterich, “Benchmarking neural network robustness to common corruptions and perturbations,” *arXiv preprint arXiv:1903.12261*, 2019.
- [17] D. Hendrycks, N. Mu, E. D. Cubuk, B. Zoph, J. Gilmer, and B. Lakshminarayanan, “Augmix: A simple data processing method to improve robustness and uncertainty,” *arXiv preprint arXiv:1912.02781*, 2019.
- [18] R. Taori, A. Dave, V. Shankar, N. Carlini, B. Recht, and L. Schmidt, “Measuring robustness to natural distribution shifts in image classification,” *Advances in neural information processing systems*, vol. 33,

- pp. 18583–18599, 2020.
- [19] D. Su, H. Zhang, H. Chen, J. Yi, P.-Y. Chen, and Y. Gao, “Is robustness the cost of accuracy?—a comprehensive study on the robustness of 18 deep image classification models,” in *European Conference on Computer Vision*, pp. 644–661, Springer, 2018.
- [20] J. Guo, W. Bao, J. Wang, Y. Ma, X. Gao, G. Xiao, A. Liu, J. Dong, X. Liu, and W. Wu, “A comprehensive evaluation framework for deep model robustness,” *Pattern Recognition*, vol. 137, p. 109308, 2023.
- [21] S. Repetto, I. Maljkovic, M. Lotto, A. E. Cinà, S. Vascon, and F. Roli, “Evaluating the robustness of explainable ai in medical image recognition under natural and adversarial data corruption,” *Machine Learning*, vol. 115, no. 1, p. 4, 2026.
- [22] S. Rahimiaghdam and H. Alemdar, “Evaluating the quality of visual explanations on chest x-ray images for thorax diseases classification,” *Neural Computing and Applications*, vol. 36, no. 17, pp. 10239–10255, 2024.
- [23] R. C. Fong and A. Vedaldi, “Interpretable explanations of black boxes by meaningful perturbation,” in *2017 IEEE international conference on computer vision (ICCV)*, pp. 3449–3457, IEEE, 2017.
- [24] R. Guidotti, “Evaluating local explanation methods on ground truth,” *Artificial Intelligence*, vol. 291, p. 103428, 2021.
- [25] I. Kim, S. Rajaraman, and S. Antani, “Visual interpretation of convolutional neural network predictions in classifying medical image modalities,” *Diagnostics*, vol. 9, no. 2, p. 38, 2019.
- [26] A. S. M. Sayeed, S. A. Birahim, M. S. Ullah, Y. Islam, A. Paul, M. A. Hasan, M. Ahsan, and M. Kowalski, “Explainable and secure federated learning for privacy-enhancing skin cancer classification using a lightweight multi-scale cnn,” *Scientific Reports*, vol. 16, no. 1, p. 11414, 2026.
- [27] A. M. Sarhan and M. A. Shaheen, “Egypli: A real-life annotated image dataset for egyptian plant leaf identification,” *Scientific Data*, 2026.
- [28] S. Islam, M. R. Ahmed, S. Islam, M. M. A. Rishad, S. Ahmed, T. R. Utshow, and M. I. Siam, “Bdmedileaves: A leaf images dataset for bangladeshi medicinal plants identification,” *Data in Brief*, vol. 50, p. 109488, 2023.
- [29] T. T. Team, “Flowers.” [http://download.tensorflow.org/example\\_images/flower\\_photos.tgz](http://download.tensorflow.org/example_images/flower_photos.tgz), jan 2019.
- [30] A. Mamaev, “Flowers recognition | kaggle,” 2018. <https://www.kaggle.com/datasets/almamaev/flowers-recognition> (accessed July 13, 2026).
- [31] S. Balasubramanian, M. A. Shah, *et al.*, “Restricted boltzmann machine with sobel filter dense adversarial noise secured layer framework for flower species recognition,” *Scientific Reports*, vol. 15, no. 1, p. 12315, 2025.
- [32] M. Toğaçar, B. Ergen, and Z. Cömert, “Classification of flower species by using features extracted from the intersection of feature selection methods in convolutional neural network models,” *Measurement*, vol. 158, p. 107703, 2020.
- [33] M. Sandler, A. Howard, M. Zhu, A. Zhmoginov, and L.-C. Chen, “Mobilenetv2: Inverted residuals and linear bottlenecks,” in *2018 IEEE/CVF conference on computer vision and pattern recognition*, pp. 4510–4520, Ieee, 2018.
- [34] G. Huang, Z. Liu, L. Van Der Maaten, and K. Q. Weinberger, “Densely connected convolutional networks,” in *2017 IEEE conference on computer vision and pattern recognition (CVPR)*, pp. 2261–2269, Ieee, 2017.
- [35] C. Szegedy, S. Ioffe, V. Vanhoucke, and A. Alemi, “Inception-v4, inception-resnet and the impact of residual connections on learning,” in *Proceedings of the AAAI conference on artificial intelligence*, vol. 31, 2017.
- [36] K. Simonyan and A. Zisserman, “Very deep convolutional networks for large-scale image recognition,” *arXiv preprint arXiv:1409.1556*, 2014.
- [37] C. Szegedy, V. Vanhoucke, S. Ioffe, J. Shlens, and Z. Wojna, “Rethinking the inception architecture for computer vision,” in *Proceedings of the IEEE conference on computer vision and pattern recognition*, pp. 2818–2826, 2016.
- [38] K. He, X. Zhang, S. Ren, and J. Sun, “Deep residual learning for image recognition,” in *Proceedings of the IEEE conference on computer vision and pattern recognition*, pp. 770–778, 2016.
- [39] T. Panboonyuen, “Seeing isn’t always believing: Analysis of grad-cam faithfulness and localization reliability in lung cancer ct classification,” *arXiv preprint arXiv:2601.12826*, 2026.
- [40] A. Chattopadhyay, A. Sarkar, P. Howlader, and V. N. Balasubramanian, “Grad-cam++: Generalized gradient-based visual explanations for deep convolutional networks,” in *2018 IEEE winter conference on applications of computer vision (WACV)*, pp. 839–847, IEEE, 2018.
- [41] P.-J. Kindermans, S. Hooker, J. Adebayo, M. Alber, K. T. Schütt, S. Dähne, D. Erhan, and B. Kim, “The (un) reliability of saliency methods,” in *Explainable AI: Interpreting, explaining and visualizing deep learning*, pp. 267–280, Springer, 2019.
- [42] H. Wang, Z. Wang, M. Du, F. Yang, Z. Zhang, S. Ding, P. Mardziel, and X. Hu, “Score-cam: Score-weighted visual explanations for convolutional neural networks,” in *2020 IEEE/CVF conference on computer vision and pattern recognition workshops (CVPRW)*, pp. 111–119, IEEE, 2020.
- [43] P.-T. Jiang, C.-B. Zhang, Q. Hou, M.-M. Cheng, and Y. Wei, “Layer-cam: Exploring hierarchical class activation maps for localization,” *IEEE transactions on image processing*, vol. 30, pp. 5875–5888, 2021.
- [44] V. Petsiuk, A. Das, and K. Saenko, “Rise: Randomized input sampling for explanation of black-box models,” *arXiv preprint arXiv:1806.07421*, 2018.
- [45] A. Nastoska, B. Jancheska, M. Rizinski, and D. Trajanov, “Evaluating trustworthiness in ai: Risks, metrics, and applications across industries,” *Electronics*, vol. 14, no. 13, p. 2717, 2025.
- [46] M. Pal, H. N. Saha, and A. Chakrabarti, “The trust-aware xai (taxai) framework: a quantitative model for interpretable and reliable clinical ai systems,” *Scientific Reports*, 2026.
- [47] S. Schneider, E. Rusak, L. Eck, O. Bringmann, W. Brendel, and M. Bethge, “Improving robustness against common corruptions by covariate shift adaptation,” *Advances in neural information processing systems*, vol. 33, pp. 11539–11551, 2020.
- [48] F. Zhang, Q. Wang, and J. Zhao, “Ar2: Attention-guided repair for the robustness of cnns against common corruptions,” *arXiv preprint arXiv:2507.06332*, 2025.
- [49] T. Mahmud, K. Barua, A. Barua, N. Basnin, S. Das, M. S. Hossain, and K. Andersson, “Explainable ai for tomato leaf disease detection: Insights into model interpretability,” in *2023 26th International Conference on Computer and Information Technology (ICCIT)*, pp. 1–6, IEEE, 2023.
- [50] B. Singh, R. P. Sharma, and S. Dey, “Interpretable plant leaf disease detection using attention-enhanced cnn,” *arXiv preprint arXiv:2512.17864*, 2025.
- [51] S. M. Shawon, F. L. Niha, O. H. Sinan, M. J. Abed, M. Al Qwaid, M. T. Sarker, and H. Zubair, “Integrating federated learning and explainable ai for plant disease detection: A systematic review of recent advances in precision agriculture,” *International Journal of Computational Intelligence Systems*, 2026.
- [52] G. Zheng, W. Ye, and A. Zhang, “Shortcutprobe: Probing prediction shortcuts for learning robust models,” *arXiv preprint arXiv:2505.13910*, 2025.
- [53] H. Baniecki, M. Chrabaszcz, A. Holzinger, B. Pfeifer, A. Saranti, and P. Biecek, “Be careful when evaluating explanations regarding ground truth,” *arXiv preprint arXiv:2311.04813*, 2023.
- [54] E. Kim, S. Kim, J. Lee, H. Kim, and S. Yoon, “Bridging the gap between classification and localization for weakly supervised object localization,” in *2022 IEEE/CVF Conference on Computer Vision and Pattern Recognition (CVPR)*, pp. 14238–14247, IEEE, 2022.



**Figure 1:** Comparison between traditional accuracy and the proposed XAI-based metrics (ETI and ERI). Although both Xception and the custom CNN achieve identical 99.45% validation accuracy, Grad-CAM heatmaps reveal that the custom CNN relies on background context. The proposed ETI and ERI metrics successfully capture this flaw by producing lower scores for the custom model.